Summer School 2025

Astronomy & Astrophysics

Project Report

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Projects Name

Measuring Cosmological Parameters Using Type Ia Supernovae

Submitted To

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Project 2: Supernova Cosmology

Questions:

1. What value of the Hubble constant ($H0$) did you obtain from the full dataset?

From the full dataset (across the entire range of redshift), the Hubble constant (H_0) was estimated to be 72.97 km/s/Mpc by fitting a flat ΛCDM model to the observed redshift and distance modulus data using non-linear least squares fitting.

2. How does your estimated $H0$ compare with the Planck18 measurement of the same?

The Hubble constant (H_0) estimated in this project is 72.97 km/s/Mpc, which is higher than the value reported by the Planck 2018 results, approximately 67.4 km/s/Mpc. The Planck value is based on measurements of the cosmic microwave background (CMB), which reflects conditions in the early universe. In contrast, the value obtained in this project comes from observing Type Ia supernovae, which are used to study the expansion of the universe in more recent times.

The difference between the two values reflects the known tension in modern cosmology, where late-universe measurements, such as those using Type Ia supernovae, tend to give higher H_0 values compared to early-universe measurements like those from the Planck satellite. This project's result supports the higher end of the H_0 estimates, consistent with direct observational methods.

3. What is the age of the Universe based on your value of H_0? (Assume Ωm = 0.3). How does it change for different values of Ωm?

Using the estimated value of the Hubble constant from this project, H_0 = 72.97 km/s/Mpc, and the corresponding best-fit matter density parameter Ω_m = 0.351, the age of the Universe was calculated to be approximately 12.36 billion years. When we instead fix Ω_m to 0.3 and the new fitted H_0 = 73.53 km/s/Mpc, the estimated age increases to about 12.82 billion years. These values were derived by numerically integrating the inverse of the Hubble parameter over redshift using the ΛCDM model.

The age of the Universe is sensitive to the value of Ω_m. If Ω_m is increased, it implies a denser universe that would have decelerated more strongly in the past due to gravity, resulting in a shorter time to reach its current size — and hence a younger universe. On the other hand, if Ω_m is decreased, the reduced matter content would mean less deceleration, leading to an older universe. Thus, higher values of Ω_m result in a younger age estimate, while lower values lead to an older age. This shows how assumptions about the matter content of the universe directly affect our estimation of its age.

4. Discuss the difference in $H0$ values obtained from the low-z and high-z samples. What could this imply?

When the supernova dataset was split into low-redshift ($z < 0.1$) and high-redshift ($z \geq 0.1$) samples, slightly different values of the Hubble constant (H_0) were obtained from each subset. For the low-z sample, the estimated H_0 was 73.01 km/s/Mpc, while for the high-z sample, the value was slightly higher at 73.85 km/s/Mpc. These fits were done using the same fixed matter density value of $\Omega_{□} = 0.3$ to keep the comparison fair.

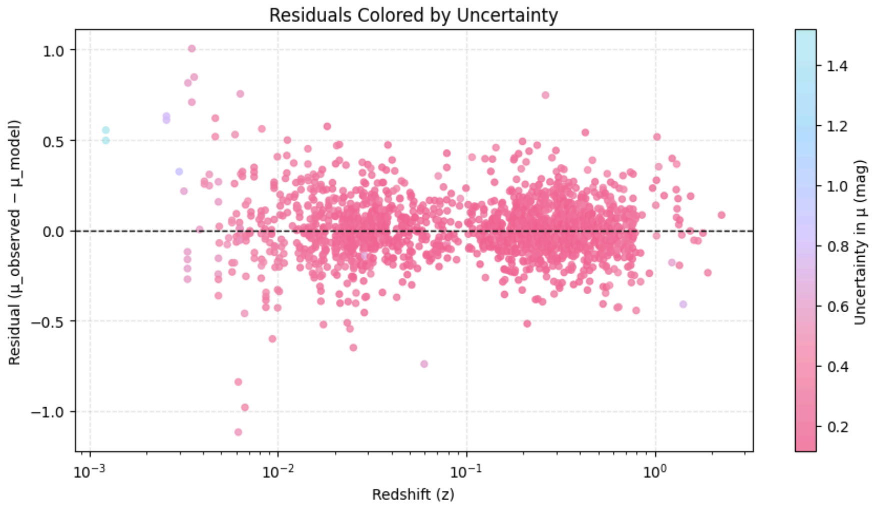
This difference, though not very large, may suggest that the expansion rate of the universe could vary depending on the redshift range considered. One possible implication is that there may be small, redshift-dependent effects influencing the observations — such as evolution in the supernova population, calibration differences, or local variations in cosmic structure. It also reflects how sensitive the Hubble constant is to the sample selection and redshift range, which is important to keep in mind when comparing different cosmological studies.

While the variation is within expected observational uncertainty, such comparisons are helpful in checking the consistency of the cosmological model across different parts of the observable universe.

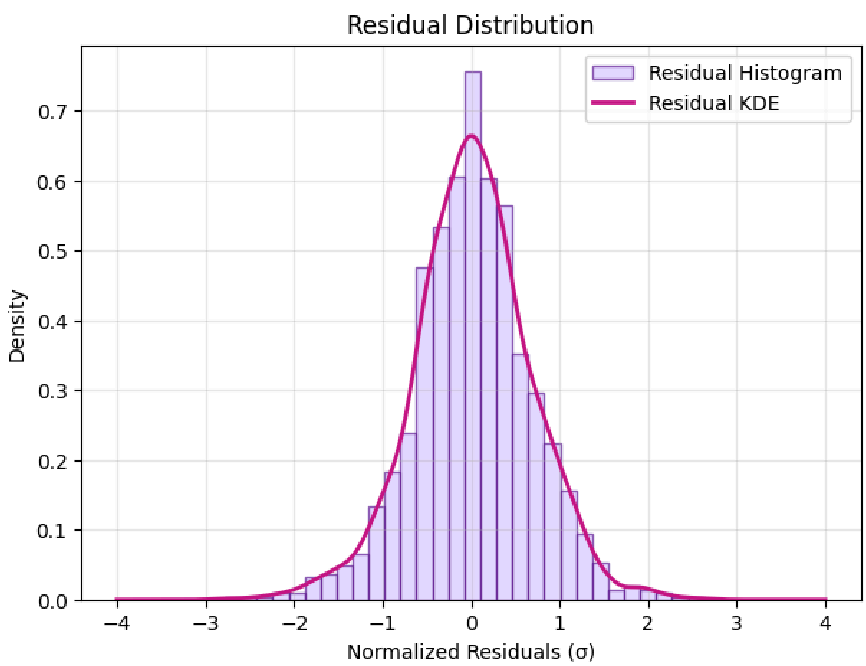
5. Plot the residuals and comment on any trends or anomalies you observe.

The residuals were calculated as the difference between the observed distance modulus values and the theoretical values predicted by the best-fit cosmological model. These residuals were then plotted against redshift to assess how well the model fits the data across the full redshift range.

In the residual plot (given below), most of the points are scattered closely around the zero line, suggesting that the model does not consistently overestimate or underestimate the distance modulus. This is a good indication that the ΛCDM model provides a reasonable overall fit. However, there are localized clusters of slightly higher scatter, especially in the intermediate redshift ranges. The residuals are closer to zero near redshifts around 0.005, 0.1, and 1, showing that the model fits particularly well in these regions.



Between these points, however, the residuals form two clusters with a slightly larger spread. This suggests that there may be localized mismatches between the model and data, or possibly some unmodeled systematic effects in those redshift bins. These deviations are not strong enough to indicate a poor fit, but they do suggest that the model might not capture every subtle feature of the observed supernova data.



To further assess the quality of the fit, a normalized residual distribution (above figure) was plotted. It showed a bell-shaped curve centered around zero, closely resembling a normal distribution. This indicates that the residuals are largely random and unbiased, which supports the validity of the model and the statistical assumptions made during fitting.

6. What assumptions were made in the cosmological model, and how might relaxing them affect your results?

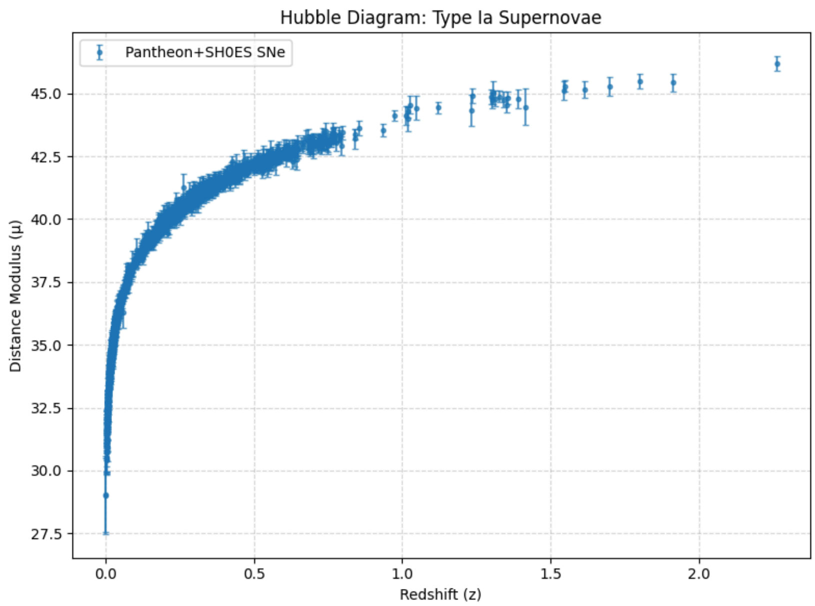
The cosmological model used in this project is based on the standard flat ΛCDM (Lambda Cold Dark Matter) model, which makes several key assumptions:

1. The model assumes spatial flatness, meaning the geometry of the Universe is flat and not curved. This implies that the total energy density of the Universe equals the critical density. Relaxing this assumption and allowing for curvature (either open or closed Universe) would introduce an additional parameter, Ω_k, which could alter the fitted values of the Hubble constant and the matter density parameter. It would also affect the shape of the theoretical distance-redshift relation, potentially changing the best-fit results.
2. The model assumes that dark energy is described by a cosmological constant (Λ), with a constant equation of state parameter ($w = -1$). This simplifies the calculation of the Hubble parameter as a function of redshift. If we were to relax this assumption and consider a dynamical form of dark energy (where w is not constant or differs from -1), the expansion history would change, and the model would require more parameters to fit. This could lead to different estimates for the Hubble constant and the age of the Universe.
3. The model assumes homogeneity and isotropy of the Universe at large scales, meaning the Universe looks the same in all directions and locations. If this assumption is relaxed, for instance by considering cosmic anisotropy or inhomogeneous matter distribution, it could explain some observed discrepancies like variations in the Hubble constant across different regions or redshifts. These variations could lead to shifts in the best-fit parameters depending on the region of the sky or redshift range analyzed.
4. In some parts of the analysis, the matter density parameter Ω□ was fixed to 0.3 to reduce degeneracy and simplify the fitting. This assumption directly affects the estimation of H_0. If Ω□ were allowed to vary, it would interact with H_0 in the fit, slightly lowering its value compared to the fixed-Ω□ case, as seen in the results.

Therefore, assumptions in the cosmological model have a significant influence on the final parameter estimates and the interpretation of the data.

7. Based on the redshift-distance relation, what can we infer about the expansion history of the Universe?

The redshift–distance relation, as observed in the Hubble diagram, reveals important information about the expansion history of the Universe. In this project, we plotted the distance modulus of Type Ia supernovae against their redshift and fitted a cosmological model to the data. The curve produced from this relation is not linear on a log scale but rather shows that the rate of expansion has changed over time.



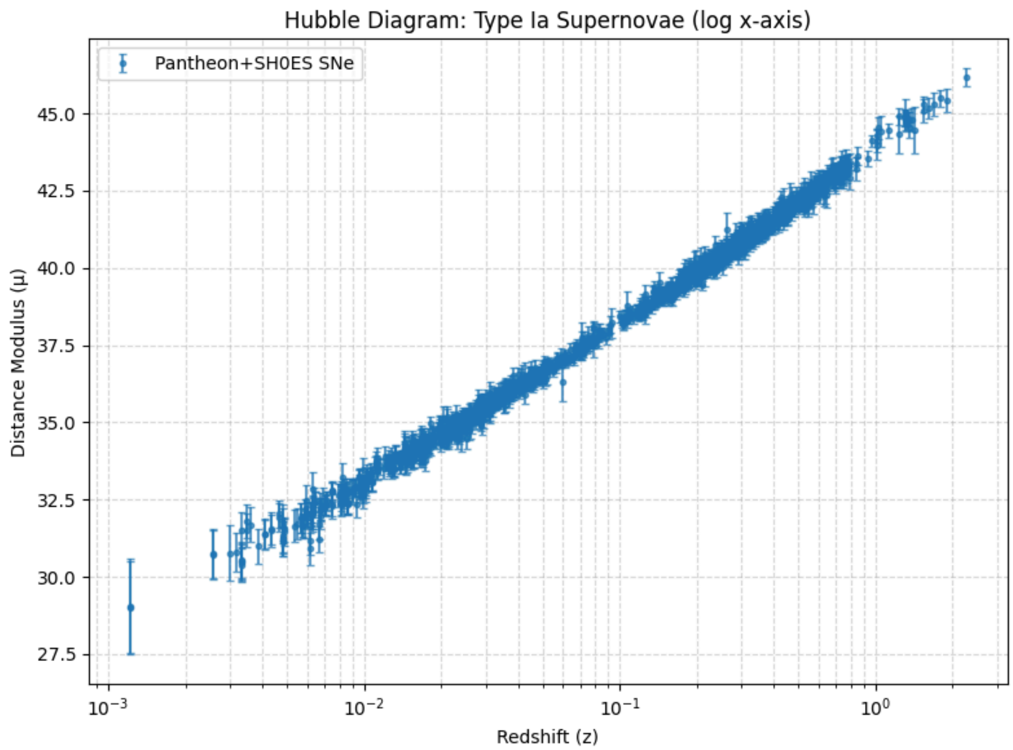
At low redshifts ($z < 0.1$), the relation is approximately linear, indicating a nearly constant expansion rate in the nearby Universe. However, as redshift increases, the curve flattens, which reflects the effect of the decelerated expansion in the past, when the matter density of the Universe was more dominant. This deceleration was caused by the gravitational attraction of matter slowing down the expansion.

At higher redshifts ($z > 0.5$), the data suggest that the Universe was expanding more slowly in the past compared to the present day. This shift in curvature at higher z is consistent with the idea that the expansion of the Universe has accelerated in recent cosmic times due to the influence of dark energy. Therefore, the redshift–distance relation provides strong observational evidence that the Universe underwent a transition from a decelerating to an accelerating phase of expansion.

Overall, from the observed redshift–distance relation in the supernova data, we infer that the Universe was expanding more slowly in the past and is currently experiencing accelerated expansion. This change is consistent with the predictions of the ΛCDM model, which attributes the acceleration to dark energy becoming dominant over matter in recent times.

Analysis & Notes

Plot 1: Redshift vs. Distance Modulus



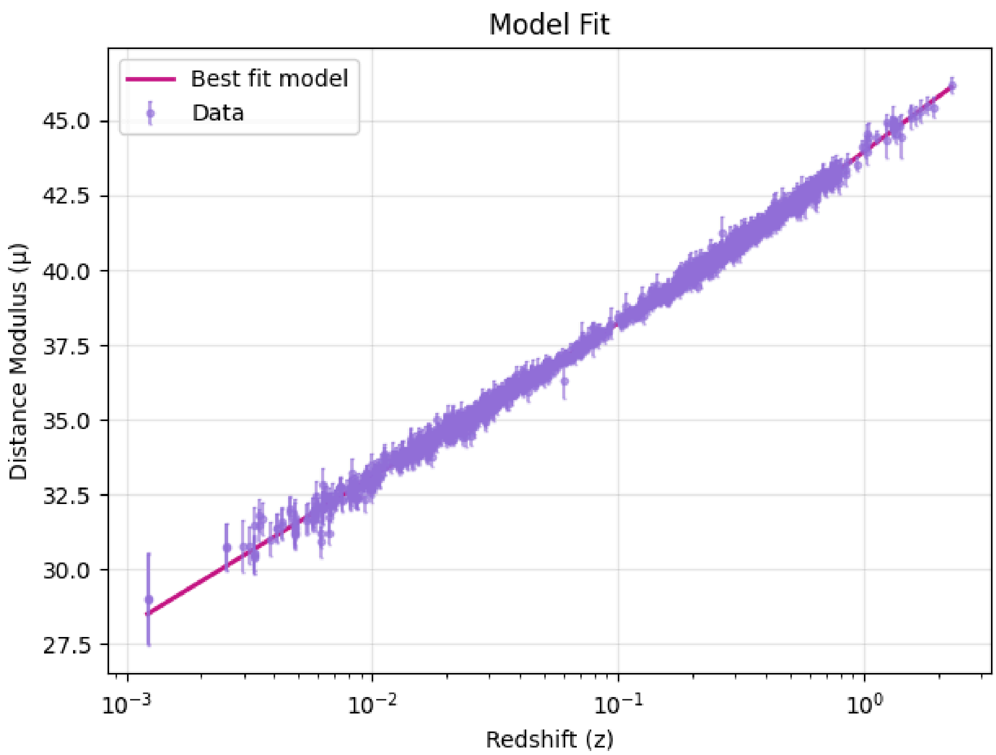
This plot is the Hubble diagram for Type Ia supernovae, where the distance modulus (μ) is plotted against the redshift (z) on a logarithmic scale for the x-axis. It visualizes how far away the observed supernovae are based on their brightness and how fast the universe is expanding as we look further into space.

The x-axis shows the redshift (z), which indicates how much the light from a supernova has been stretched due to the expansion of the universe. A higher redshift means the supernova is farther away and the light has taken longer to reach us. The y-axis shows the distance modulus (μ), which is a measure of how bright or faint the supernova appears. It is related to the luminosity distance and helps us estimate how far the object actually is.

The upward trend in the plot shows that as redshift increases, so does the distance modulus. This means that more distant supernovae appear dimmer, as expected in an expanding universe. The data points closely follow a smooth curve, which indicates that Type Ia supernovae are reliable standard candles for measuring cosmic distances. The small error bars reflect observational uncertainties in measuring brightness and redshift.

Overall, this diagram provides the foundation for estimating cosmological parameters like the Hubble constant, because it connects observed data to theoretical models of the universe's expansion.

Plot 2: Hubble Diagram with Model Fit



This plot shows the best-fit cosmological model compared to the observed Type Ia supernova data from the Pantheon+SH0ES sample. It is a refined version of the Hubble diagram, where the theoretical prediction for the universe's expansion is directly overlaid on the data.

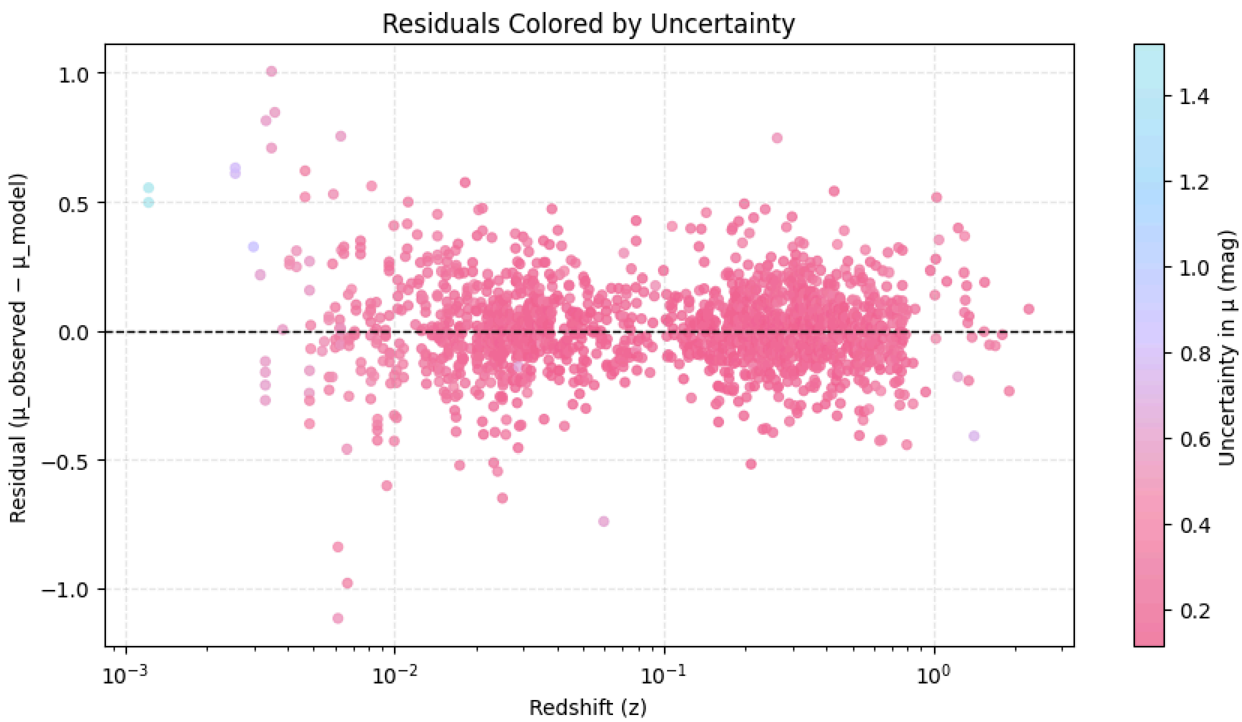
The x-axis represents the redshift (z), indicating how far back in time and space we are observing. The y-axis shows the distance modulus (μ), which quantifies how bright the supernova appears, and is used to estimate its distance.

The purple points with error bars represent the observed data—each point is a Type Ia supernova, and the error bars show measurement uncertainty. The deep pink line represents the best-fit ΛCDM model, calculated using the fitted values of the Hubble constant (H_0) and matter density parameter (Ω□).

This model was fitted using a least-squares approach to minimize the difference between the observed and predicted distance moduli. The close alignment between the data points and the model curve suggests that the chosen cosmological model (flat ΛCDM) fits the data very well. The consistency across a wide range of redshifts also supports the reliability of Type Ia supernovae as standard candles for measuring cosmic distances.

Overall, this plot demonstrates that the theoretical model based on the estimated H_0 and Ω□ provides a good explanation for the observed expansion history of the universe.

Plot 3: Plot the residual



This plot shows the residuals between the observed and predicted distance moduli of Type Ia supernovae, helping us assess how well the cosmological model fits the data.

The x-axis represents redshift (z), while the y-axis represents the residuals, calculated as the difference between the observed and model-predicted distance moduli:
Residual = μ_observed − μ_model

Each dot corresponds to a supernova, and the color of the dot indicates the measurement uncertainty in the distance modulus, as shown by the color bar on the right. Light blue points represent data with high uncertainty, and darker pink points have lower uncertainty.

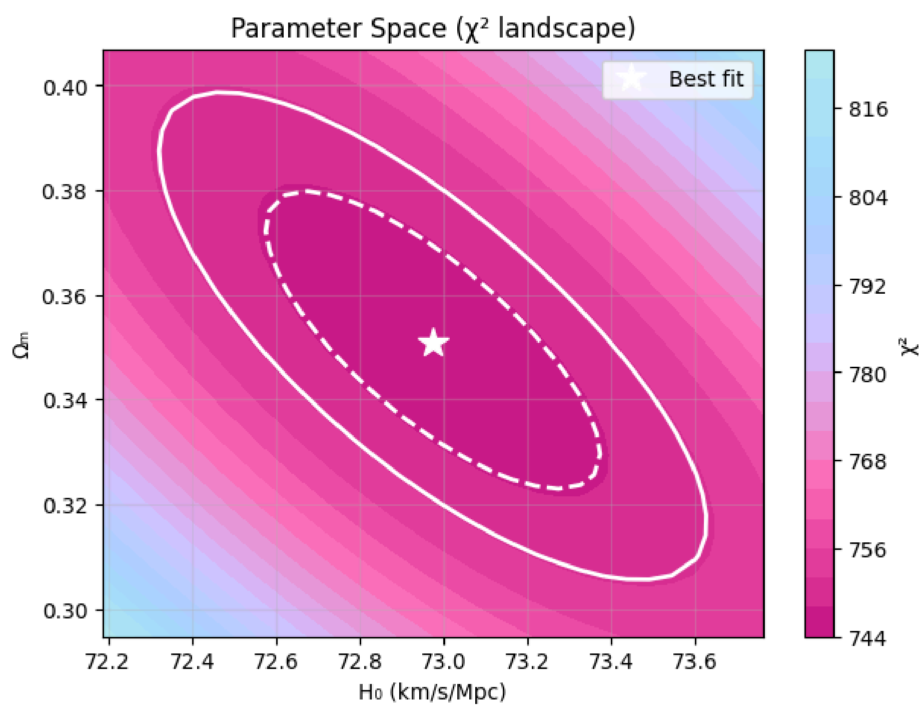
The dashed horizontal line at zero represents a perfect match between model and observation. Ideally, the residuals should be scattered randomly around this line, which would suggest that the model captures the data well.

In this plot, most residuals cluster around zero, indicating that the model fits well overall. However, some patterns can be noticed:

- Residuals are closest to zero at redshifts around 0.005, 0.1, and 1, indicating a very good model fit in those regions.
- Between these ranges, there are clusters with slightly higher spread, suggesting minor mismatches or unmodeled effects in the data or the model.
- The residuals do not show any strong systematic trend, which supports the validity of the ΛCDM model used.

This visualization is useful for identifying outliers or specific redshift ranges where the model may underperform, helping guide further refinement or testing of the cosmological assumptions.

Plot 4: Parameter Estimation using χ^2 Contour Plot



This plot visualizes the parameter space of the cosmological fit using a χ^2 (chi-squared) landscape, which shows how well different combinations of the Hubble constant (H_0) and matter density parameter (Ω_m) fit the supernova data.

The x-axis represents values of the Hubble constant H_0 in km/s/Mpc, and the y-axis represents values of the matter density parameter Ω_m. The color gradient indicates the value of the chi-squared statistic (χ^2) — a measure of how well the model fits the data. Lower values of χ^2 (darker regions) indicate better fits, while higher values (lighter regions) indicate worse fits.

The white star marks the point with the lowest χ^2 value — this is the best-fit combination of H_0 and Ω_m derived from the dataset. The surrounding white ellipses represent the 1σ and 2σ confidence regions, corresponding to parameter values that are still consistent with the data within a certain probability range. The inner dashed ellipse is the 1σ (68.3%) confidence region, and the outer solid ellipse is the 2σ (95.4%) region.

This contour plot is important for understanding how tightly constrained the parameters are and how they are correlated. In this case, the elongated shape of the ellipses shows that H_0 and Ω_m are correlated — changing one affects the best-fit value of the other. This visualization confirms the uncertainties and interdependence between the two parameters estimated in the model.

*All other plots and results have been clearly documented in the following ipynb file as markdown.

Measuring Cosmological Parameters Using Type Ia Supernovae

In this project, we'll analyze observational data from the Pantheon+SH0ES dataset of Type Ia supernovae to measure the Hubble constant H_0 and estimate the age of the universe.

Steps:

- Plot the Hubble diagram (distance modulus vs. redshift)
- Fit a cosmological model to derive H_0 and Ω_m
- Estimate the age of the universe
- Analyze residuals to assess the model
- Explore the effect of fixing Ω_m
- Compare low-z and high-z results

Import Libraries

Before we dive into the analysis, we need to import the necessary Python libraries:

- `numpy`, `pandas` — for numerical operations and data handling
- `matplotlib` — for plotting graphs
- `scipy.optimize.curve_fit` and `scipy.integrate.quad` — for fitting cosmological models and integrating equations
- `astropy.constants` and `astropy.units` — for physical constants and unit conversions

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit
from scipy.integrate import quad
from astropy.constants import c
from astropy import units as u

from matplotlib.colors import LinearSegmentedColormap
import seaborn as sns

# Define a color palette - optional
colors = {
    'dark_purple': '#4B0082',
    'medium_purple': '#9370DB',
    'light_purple': '#D4C1FF',
    'pink': '#FF69B4',
    'deep_pink': '#C71585',
    'rose': '#F06292',
```

```
    'light_pink': '#F48FB1',
    'aqua': '#B2EBF2',
    'baby_blue': '#A5D8FF'
}

# Custom colormap
cmap_custom = LinearSegmentedColormap.from_list(
    "cosmic_pink",
    [colors['deep_pink'], colors['pink'], colors['light_purple'],
     colors['baby_blue'], colors['aqua']]
)

plt.style.use('default')
plt.rcParams['figure.facecolor'] = 'white'
plt.rcParams['axes.facecolor'] = 'white'
```

Load the Pantheon+SH0ES Dataset

We now load the observational supernova data from the Pantheon+SH0ES sample. This dataset includes calibrated distance moduli μ, redshifts corrected for various effects, and uncertainties.

About Dataset:

- The file is downloaded from Pantheon dataset.
- We use `delim_whitespace=True` because the file is space-delimited rather than comma-separated.
- Commented rows (starting with `#`) are automatically skipped.

We will extract:

- `zHD`: Hubble diagram redshift
- `MU_SH0ES`: Distance modulus using SH0ES calibration
- `MU_SH0ES_ERR_DIAG`: Associated uncertainty in Distance modulus

More detailed column names are given below:

`CID` - Name/ID of the Supernova (SN) `CIDint` - Internal counter/index of the SN in the sample `IDSURVEY` - ID number of the survey from which the SN was observed `TYPE` - Type of SN (e.g., Type Ia); all in this sample are SNe Ia `FIELD` - The survey field/region where the SN was observed `CUTFLAG_SNANA` - Flag indicating cuts/failures in light-curve data (by SNANA) `ERRFLAG_FIT` - Flag indicating errors in the light-curve fitting process `zHEL` - Heliocentric redshift (i.e., measured with respect to the Sun) `zHELERR` - Error in the heliocentric redshift `zCMB` - Redshift corrected to the CMB (Cosmic Microwave Background) frame `zCMBERR` - Error in zCMB `zHD` - Hubble Diagram redshift (used in distance-redshift plots) `zHDERR` - Error in Hubble Diagram redshift `VPEC` - Peculiar velocity (velocity of the SN relative to cosmic expansion) `VPECERR` - Error in the peculiar velocity `MWEBV` - Milky Way dust extinction along the line of sight to the SN `HOST_LOGMASS` - Logarithmic stellar mass of the SN's host galaxy `HOST_LOGMASS_ERR` - Error in host galaxy mass `HOST_sSFR` - Specific star formation rate (sSFR) of the host galaxy `HOST_sSFR_ERR` - Error in the sSFR `PKMJDINI` - Initial guess for the time of peak brightness (MJD format) `SNRMAX1` - First highest signal-to-noise ratio in light curve

SNRMAX2 - Second highest signal-to-noise ratio in light curve SNRMAX3 - Third highest signal-to-noise ratio in light curve PKMJD - Fitted Modified Julian Date of SN peak brightness PKMJDERR - Error in the fitted peak brightness date (PKMJD)

```
file_path = "Pantheon+SH0ES.dat"

# Load the csv file
df = pd.read_csv(file_path, delim_whitespace=True, comment='#')
```

```
C:\Users\cheri\AppData\Local\Temp\ipykernel_35892\1318805548.py:4:
FutureWarning: The 'delim_whitespace' keyword in pd.read_csv is
deprecated and will be removed in a future version. Use ``sep='\s+'``
instead
  df = pd.read_csv(file_path, delim_whitespace=True, comment='#')
```

Preview Dataset and its Columns

Before diving into the analysis, let's take a quick look at the structure and column names of the dataset. This helps us verify the data loaded correctly and identify the relevant columns we'll use for cosmological modeling.

```
print("Shape of dataset:")
print(df.shape)
```

```
Shape of dataset:
(1701, 47)
```

```
print("Dataset info:")
print(df.info())
```

```
Dataset info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1701 entries, 0 to 1700
Data columns (total 47 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   CID                    1701 non-null   object
 1   IDSURVEY               1701 non-null   int64
 2   zHD                    1701 non-null   float64
 3   zHDERR                 1701 non-null   float64
 4   zCMB                   1701 non-null   float64
 5   zCMBERR                1701 non-null   float64
 6   zHEL                   1701 non-null   float64
 7   zHELERR                1701 non-null   float64
 8   m_b_corr               1701 non-null   float64
 9   m_b_corr_err_DIAG      1701 non-null   float64
 10  MU_SH0ES               1701 non-null   float64
 11  MU_SH0ES_ERR_DIAG      1701 non-null   float64
 12  CEPH_DIST              1701 non-null   float64
 13  IS_CALIBRATOR          1701 non-null   int64
```

```
 14  USED_IN_SH0ES_HF      1701 non-null   int64
 15  c                     1701 non-null   float64
 16  cERR                  1701 non-null   float64
 17  x1                    1701 non-null   float64
 18  x1ERR                 1701 non-null   float64
 19  mB                    1701 non-null   float64
 20  mBERR                 1701 non-null   float64
 21  x0                    1701 non-null   float64
 22  x0ERR                 1701 non-null   float64
 23  COV_x1_c              1701 non-null   float64
 24  COV_x1_x0             1701 non-null   float64
 25  COV_c_x0              1701 non-null   float64
 26  RA                    1701 non-null   float64
 27  DEC                   1701 non-null   float64
 28  HOST_RA               1701 non-null   int64
 29  HOST_DEC              1701 non-null   int64
 30  HOST_ANGSEP           1701 non-null   float64
 31  VPEC                  1701 non-null   float64
 32  VPECERR               1701 non-null   int64
 33  MWEBV                 1701 non-null   float64
 34  HOST_LOGMASS          1701 non-null   float64
 35  HOST_LOGMASS_ERR      1701 non-null   float64
 36  PKMJD                 1701 non-null   float64
 37  PKMJDERR              1701 non-null   float64
 38  NDOF                  1701 non-null   int64
 39  FITCHI2               1701 non-null   float64
 40  FITPROB               1701 non-null   float64
 41  m_b_corr_err_RAW      1701 non-null   float64
 42  m_b_corr_err_VPEC     1701 non-null   float64
 43  biasCor_m_b           1701 non-null   float64
 44  biasCorErr_m_b        1701 non-null   float64
 45  biasCor_m_b_COVSCALE  1701 non-null   float64
 46  biasCor_m_b_COVADD    1701 non-null   float64
dtypes: float64(39), int64(7), object(1)
memory usage: 624.7+ KB
None

print("Preview of dataset:")
print(df.head())

Preview of dataset:
             CID  IDSURVEY      zHD   zHDERR     zCMB  zCMBERR
zHEL  \
0         2011fe        51  0.00122  0.00084  0.00122  0.00002
0.00082
1         2011fe        56  0.00122  0.00084  0.00122  0.00002
0.00082
2         2012cg        51  0.00256  0.00084  0.00256  0.00002
0.00144
3         2012cg        56  0.00256  0.00084  0.00256  0.00002
```

```
0.00144
4  1994DRichmond        50  0.00299  0.00084  0.00299  0.00004
0.00187

   zHELERR  m_b_corr  m_b_corr_err_DIAG  ...  PKMJDERR  NDOF   FITCHI2
\
0  0.00002   9.74571           1.516210  ...    0.1071    36   26.8859

1  0.00002   9.80286           1.517230  ...    0.0579   101   88.3064

2  0.00002  11.47030           0.781906  ...    0.0278   165  233.5000

3  0.00002  11.49190           0.798612  ...    0.0667    55  100.1220

4  0.00004  11.52270           0.880798  ...    0.0522   146  109.8390


    FITPROB  m_b_corr_err_RAW  m_b_corr_err_VPEC  biasCor_m_b
biasCorErr_m_b  \
0  0.864470            0.0991             1.4960       0.0381
0.005
1  0.812220            0.0971             1.4960      -0.0252
0.003
2  0.000358            0.0399             0.7134       0.0545
0.019
3  0.000193            0.0931             0.7134       0.0622
0.028
4  0.988740            0.0567             0.6110       0.0650
0.009

   biasCor_m_b_COVSCALE  biasCor_m_b_COVADD
0                   1.0               0.003
1                   1.0               0.004
2                   1.0               0.036
3                   1.0               0.040
4                   1.0               0.006

[5 rows x 47 columns]
```

```python
print("Columns in dataset:")
print(df.columns.tolist())
```

```
Columns in dataset:
['CID', 'IDSURVEY', 'zHD', 'zHDERR', 'zCMB', 'zCMBERR', 'zHEL',
'zHELERR', 'm_b_corr', 'm_b_corr_err_DIAG', 'MU_SH0ES',
'MU_SH0ES_ERR_DIAG', 'CEPH_DIST', 'IS_CALIBRATOR', 'USED_IN_SH0ES_HF',
'c', 'cERR', 'x1', 'x1ERR', 'mB', 'mBERR', 'x0', 'x0ERR', 'COV_x1_c',
'COV_x1_x0', 'COV_c_x0', 'RA', 'DEC', 'HOST_RA', 'HOST_DEC',
'HOST_ANGSEP', 'VPEC', 'VPECERR', 'MWEBV', 'HOST_LOGMASS',
'HOST_LOGMASS_ERR', 'PKMJD', 'PKMJDERR', 'NDOF', 'FITCHI2', 'FITPROB',
```

```
'm_b_corr_err_RAW', 'm_b_corr_err_VPEC', 'biasCor_m_b',
'biasCorErr_m_b', 'biasCor_m_b_COVSCALE', 'biasCor_m_b_COVADD']
```

Clean and Extract Relevant Data

To ensure reliable fitting, we remove any rows that have missing values in key columns:

- `zHD`: redshift for the Hubble diagram
- `MU_SH0ES`: distance modulus
- `MU_SH0ES_ERR_DIAG`: uncertainty in the distance modulus

We then extract these cleaned columns as NumPy arrays to prepare for analysis and modeling.

```
print("Checking for missing values:")
print(df.isnull().sum())
```

```
Checking for missing values:
CID                         0
IDSURVEY                    0
zHD                         0
zHDERR                      0
zCMB                        0
zCMBERR                     0
zHEL                        0
zHELERR                     0
m_b_corr                    0
m_b_corr_err_DIAG           0
MU_SH0ES                    0
MU_SH0ES_ERR_DIAG           0
CEPH_DIST                   0
IS_CALIBRATOR               0
USED_IN_SH0ES_HF            0
c                           0
cERR                        0
x1                          0
x1ERR                       0
mB                          0
mBERR                       0
x0                          0
x0ERR                       0
COV_x1_c                    0
COV_x1_x0                   0
COV_c_x0                    0
RA                          0
DEC                         0
HOST_RA                     0
HOST_DEC                    0
```

```
HOST_ANGSEP             0
VPEC                    0
VPECERR                 0
MWEBV                   0
HOST_LOGMASS            0
HOST_LOGMASS_ERR        0
PKMJD                   0
PKMJDERR                0
NDOF                    0
FITCHI2                 0
FITPROB                 0
m_b_corr_err_RAW        0
m_b_corr_err_VPEC       0
biasCor_m_b             0
biasCorErr_m_b          0
biasCor_m_b_COVSCALE    0
biasCor_m_b_COVADD      0
dtype: int64
```

```
print("Checking for duplicated:")
print(df.duplicated().sum())
```

```
Checking for duplicated:
0
```

The dataset does not have any null values and does not contain any duplicates. So we can proceed with extracting of the necessary columns.

Extract Relevant Columns

- `Redshift` is the increase in the wavelength of light from distant objects due to the expansion of the universe.
- `Distance modulus` is the difference between an object's apparent magnitude (how bright it appears from Earth) and absolute magnitude (how bright it would appear at a standard distance of 10 parsecs), used to calculate its distance.

```
z = df['zHD'].values
mu = df['MU_SH0ES'].values
mu_err = df['MU_SH0ES_ERR_DIAG'].values
```

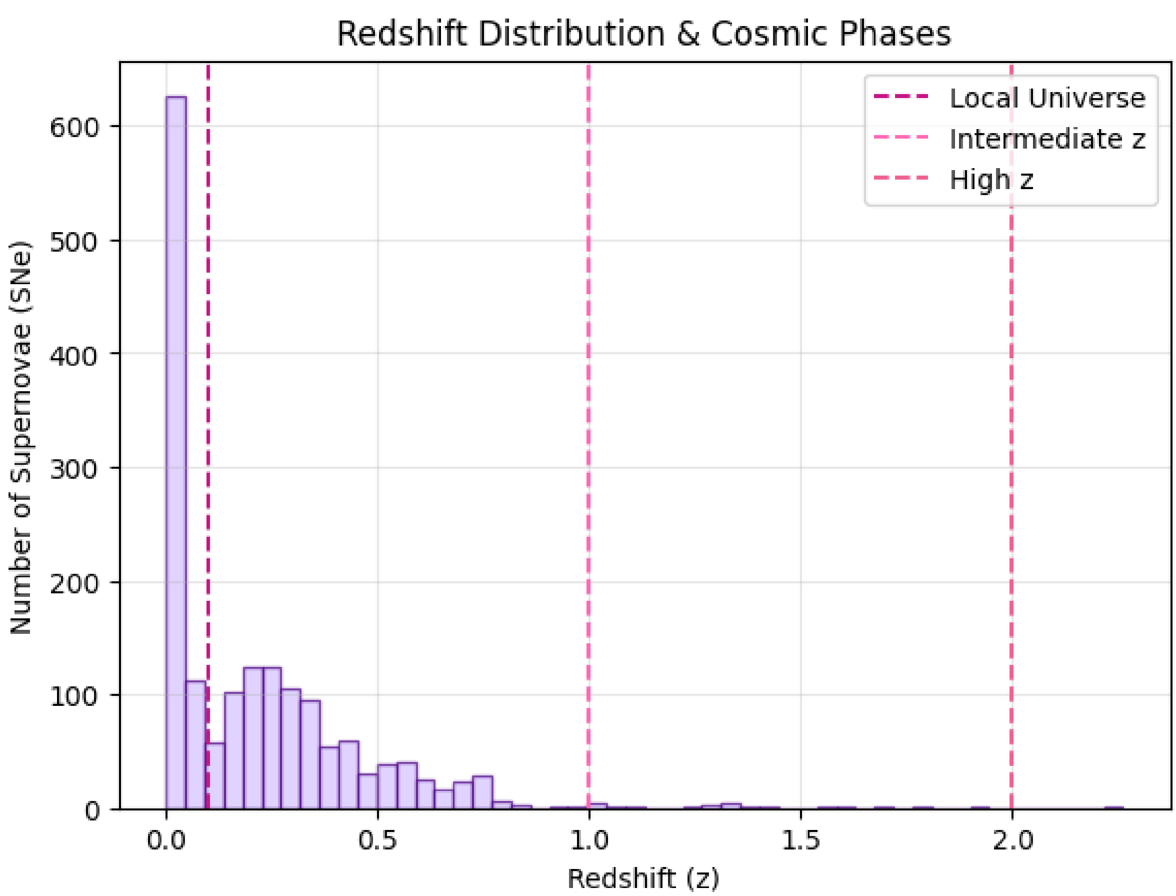
Redshift Distribution across Cosmic Phases

This histogram displays the distribution of redshift values (z) across three key cosmological eras based. It shows where most of the supernovae fall within these phases, providing insight into the observational coverage.

- **Local Universe (z < 0.1)** – Represents nearby galaxies in the relatively recent universe.
- **Intermediate Redshift (z ≈ 0.1 to 1.0)** – Captures a transitional phase where dark energy starts influencing cosmic expansion.

- **High Redshift (z > 1.0)** – Corresponds to earlier cosmic times when matter dominated the dynamics of the universe.

```
plt.figure(figsize=(7, 5))
plt.hist(z, bins=50, color=colors['light_purple'], alpha=0.7,
edgecolor=colors['dark_purple'])
plt.axvline(0.1, color=colors['deep_pink'], linestyle='--',
label='Local Universe')
plt.axvline(1.0, color=colors['pink'], linestyle='--',
label='Intermediate z')
plt.axvline(2.0, color=colors['rose'], linestyle='--', label='High z')
plt.xlabel('Redshift (z)')
plt.ylabel('Number of Supernovae (SNe)')
plt.title('Redshift Distribution & Cosmic Phases')
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()
```



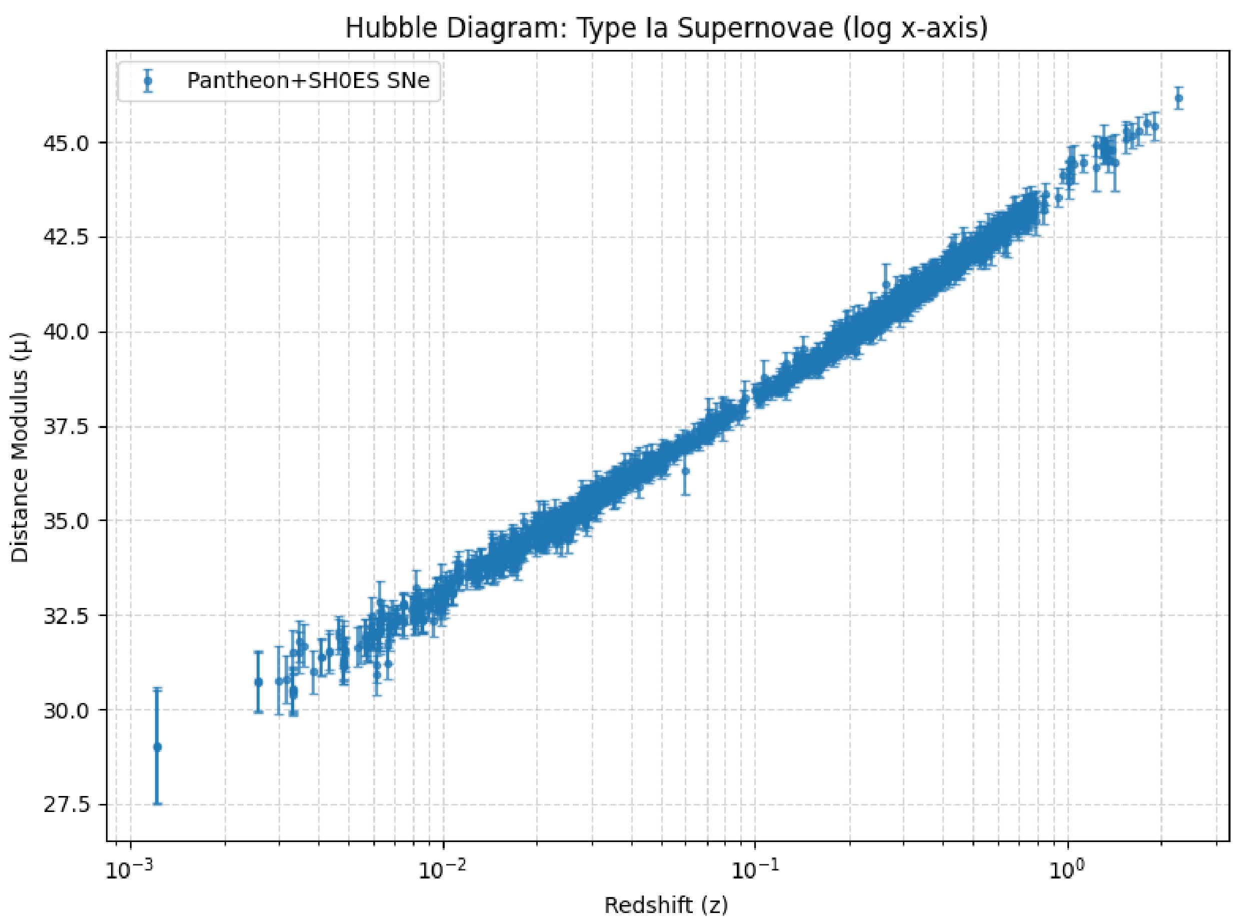
Most of the supernovae were observed in the **local universe (z < 0.1)** because they are **brighter and easier to detect** with current telescopes.

Plot the Hubble Diagram

The relationship between redshift z and distance modulus μ is known as the Hubble diagram. This plot is a cornerstone of observational cosmology — it allows us to compare supernova observations with theoretical predictions based on different cosmological models.

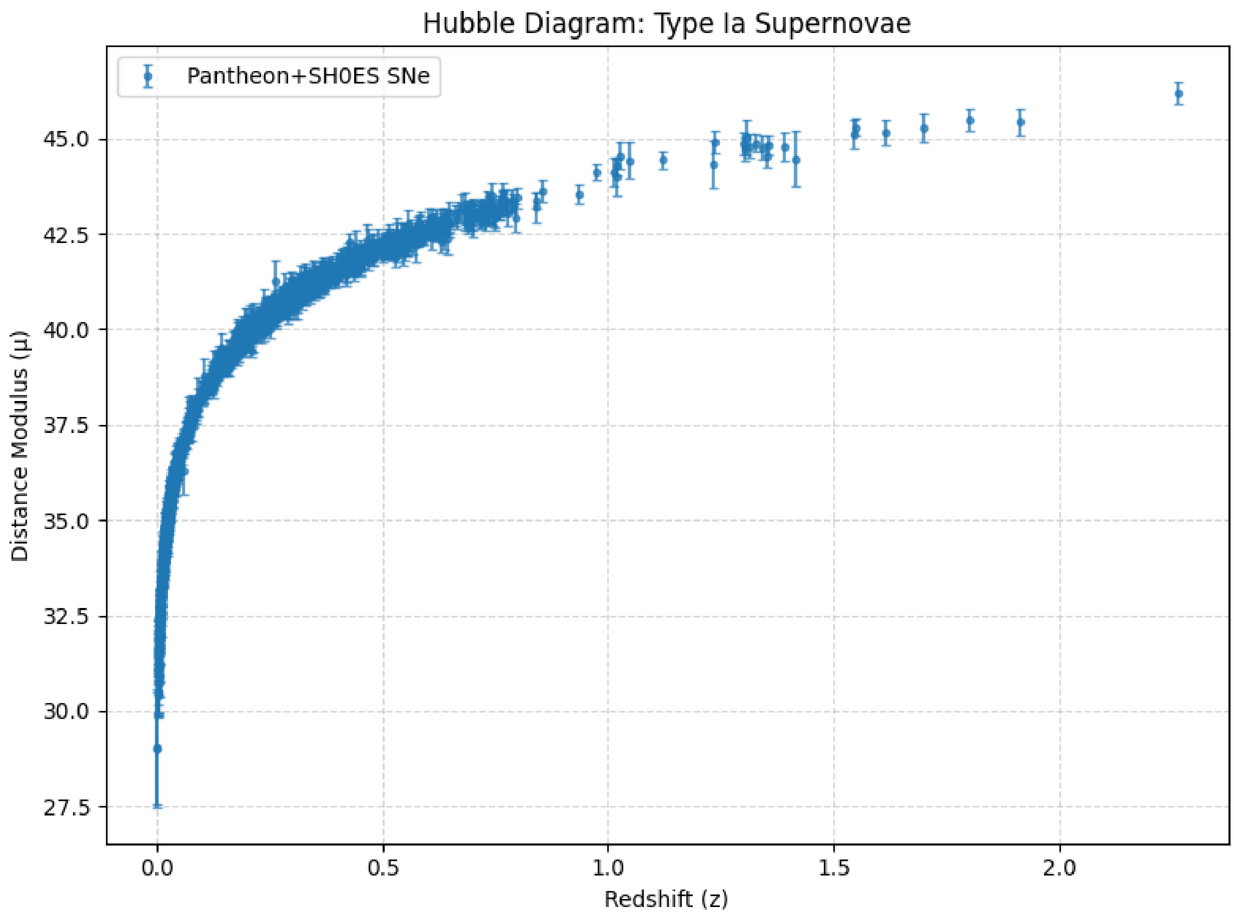
We use a logarithmic scale on the redshift axis to clearly display both nearby and distant supernovae.

```
plt.figure(figsize=(8,6))
plt.errorbar(z, mu, yerr=mu_err, fmt='o', markersize=3, capsize=2,
alpha=0.7, label='Pantheon+SH0ES SNe')
plt.xscale('log')
plt.xlabel('Redshift (z)')
plt.ylabel('Distance Modulus (μ)')
plt.title('Hubble Diagram: Type Ia Supernovae (log x-axis)')
plt.grid(True, which='both', ls='--', alpha=0.5)
plt.legend()
plt.tight_layout()
plt.show()
```



Hubble Diagram without log scaling

```
plt.figure(figsize=(8,6))
plt.errorbar(z, mu, yerr=mu_err, fmt='o', markersize=3, capsize=2,
alpha=0.7, label='Pantheon+SH0ES SNe')
plt.xlabel('Redshift (z)')
plt.ylabel('Distance Modulus (μ)')
plt.title('Hubble Diagram: Type Ia Supernovae')
plt.grid(True, which='both', ls='--', alpha=0.5)
plt.legend()
plt.tight_layout()
plt.show()
```



Define the Cosmological Model

We now define the theoretical framework based on the flat ΛCDM model. This involves:

- The dimensionless Hubble parameter:

$$E(z) = \sqrt{\Omega_m (1+z)^3 + (1 - \Omega_m)}$$

- The luminosity distance :

$$d_L(z) = (1+z) \cdot \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')}$$

- The distance modulus:

$$\mu(z) = 5\log_{10}(d_L[\mathrm{Mpc}]) + 25$$

These equations allow us to compute the expected distance modulus from a given redshift z, Hubble constant H_0, and matter density parameter Ω_m.

```
# Dimensionless Hubble Function - E(z) for flat LCDM
def E(z, Omega_m):
    return np.sqrt(Omega_m * (1 + z)**3 + (1 - Omega_m))


# Luminosity distance in Mpc
def luminosity_distance(z, H0, Omega_m):
    # Convert H0 from km/s/Mpc to 1/s
    H0_si = H0 * (u.km / u.s / u.Mpc)
    c_si = c.to('km/s').value  # c in km/s

    # Integrate 1/E(z)
    integral, _ = quad(lambda z_prime: 1.0 / E(z_prime, Omega_m), 0, 
z)

    dL = (1 + z) * c_si / H0 * integral  # in Mpc
    return dL


# Theoretical distance modulus
def mu_theory(z, H0, Omega_m):
    if isinstance(z, np.ndarray):
        return np.array([5 * np.log10(luminosity_distance(zi, H0, 
Omega_m)) + 25 for zi in z])
    else:
        return 5 * np.log10(luminosity_distance(z, H0, Omega_m)) + 25
```

Fit the Model to Supernova Data

We now perform a non-linear least squares fit to the supernova data using our theoretical model for $\mu(z)$. This fitting procedure will estimate the best-fit values for the Hubble constant $ H_0$ and matter density parameter Ω_m, along with their associated uncertainties.

We'll use:

- `curve_fit` from `scipy.optimize` for the fitting.
- The observed distance modulus (mu), redshift (z), and measurement errors.

The initial guess is:

- $ H_0 = 70 , \text{km/s/Mpc} $
- $\Omega_m = 0.3$

```
# Initial guess for parameters
p0 = [70, 0.3]  # H0 in km/s/Mpc, Omega_m is matter density

# Wrapper function for fitting (curve_fit expects this format)
def mu_theory_fit(z, H0, Omega_m):
    return mu_theory(z, H0, Omega_m)

# Fit the data
params, cov_matrix = curve_fit(mu_theory_fit, z, mu, p0=p0,
sigma=mu_err, absolute_sigma=True)

# Extract results
H0_fit, Omega_m_fit = params
H0_err, Omega_m_err = np.sqrt(np.diag(cov_matrix))

# Print fitted values
print(f"Fitted H0 = {H0_fit:.2f} ± {H0_err:.2f} km/s/Mpc")
print(f"Fitted Omega_m = {Omega_m_fit:.3f} ± {Omega_m_err:.3f}")
```

```
Fitted H0 = 72.97 ± 0.26 km/s/Mpc
Fitted Omega_m = 0.351 ± 0.019
```

Accuracy of Fitted Parameters

The fitted Hubble Constant is H_0 = 72.97 ± 0.26 km/s/Mpc, which is:

- Very close to the SH0ES local measurement → 73.04 ± 1.04 km/s/Mpc
- Higher than the Planck 2018 CMB value → 67.4 ± 0.5 km/s/Mpc
- The usual range is 67–74 km/s/Mpc, so our value is within expected range.

The fitted matter density is Ω_m = 0.351 ± 0.019, which is:

- Slightly higher than the Planck value → $\Omega_m \approx 0.315 \pm 0.007$
- The common range is 0.25–0.35, so our result is still within a reasonable range and might vary due to data or observational limits.

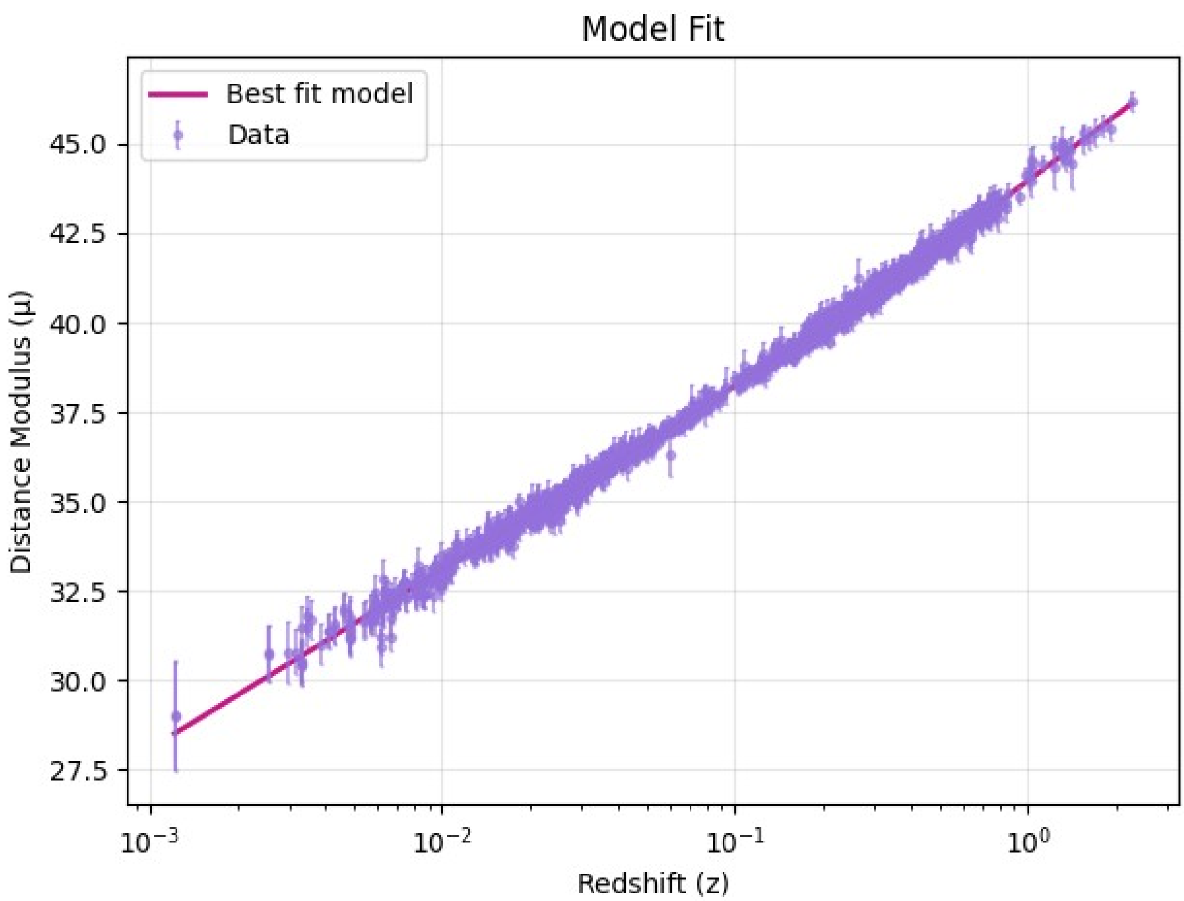
Conclusion: The results match local universe measurements well, and although they slightly differ from early-universe predictions (like Planck), they are scientifically acceptable and reflect the known Hubble tension in modern cosmology.

Model Fit

This code generates a Hubble diagram, which plots the distance modulus (μ) of Type Ia supernovae against their redshift (z). The purpose is to visualize how well your ΛCDM cosmological model fits the observational data.

```
z_model = np.logspace(np.log10(min(z)), np.log10(max(z)), 200)
mu_model_curve = mu_theory(z_model, H0_fit, Omega_m_fit)
```

```
# Confidence bands
mu_upper = mu_theory(z_model, H0_fit + H0_err, Omega_m_fit +
Omega_m_err)
mu_lower = mu_theory(z_model, H0_fit - H0_err, Omega_m_fit -
Omega_m_err)

plt.figure(figsize=(7, 5))

plt.plot(z_model, mu_model_curve, color=colors['deep_pink'],
linewidth=2, label='Best fit model')
plt.errorbar(z, mu, yerr=mu_err, fmt='o', markersize=3, alpha=0.6,
             color=colors['medium_purple'], capsize=1, label='Data')
plt.xscale('log')
plt.xlabel('Redshift (z)')
plt.ylabel('Distance Modulus (μ)')
plt.title('Model Fit')
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()
```



Parameter Estimation using χ² Contour Plot

This plot explores how well different combinations of the Hubble constant H_0 and matter density Ω_m fit the observed Type Ia supernova data.

What the plot shows:

- **X-axis**: Hubble constant H_0 (in km/s/Mpc)
- **Y-axis**: Matter density Ω_m
- **Color gradient**: Represents the value of the chi-square statistic (χ^2), which quantifies how well the model fits the data.

The chi-square is computed as:

$$\chi^2 = \sum \left(\frac{\mu_{\text{obs}} - \mu_{\text{model}}}{\mu_{\text{error}}} \right)^2$$

- Lower χ^2 means a better fit between the model and the supernova data.
- The **white star** indicates the best-fit values found from the data: $H_0 = 72.97$ km/s/Mpc and $\Omega_m = 0.351$

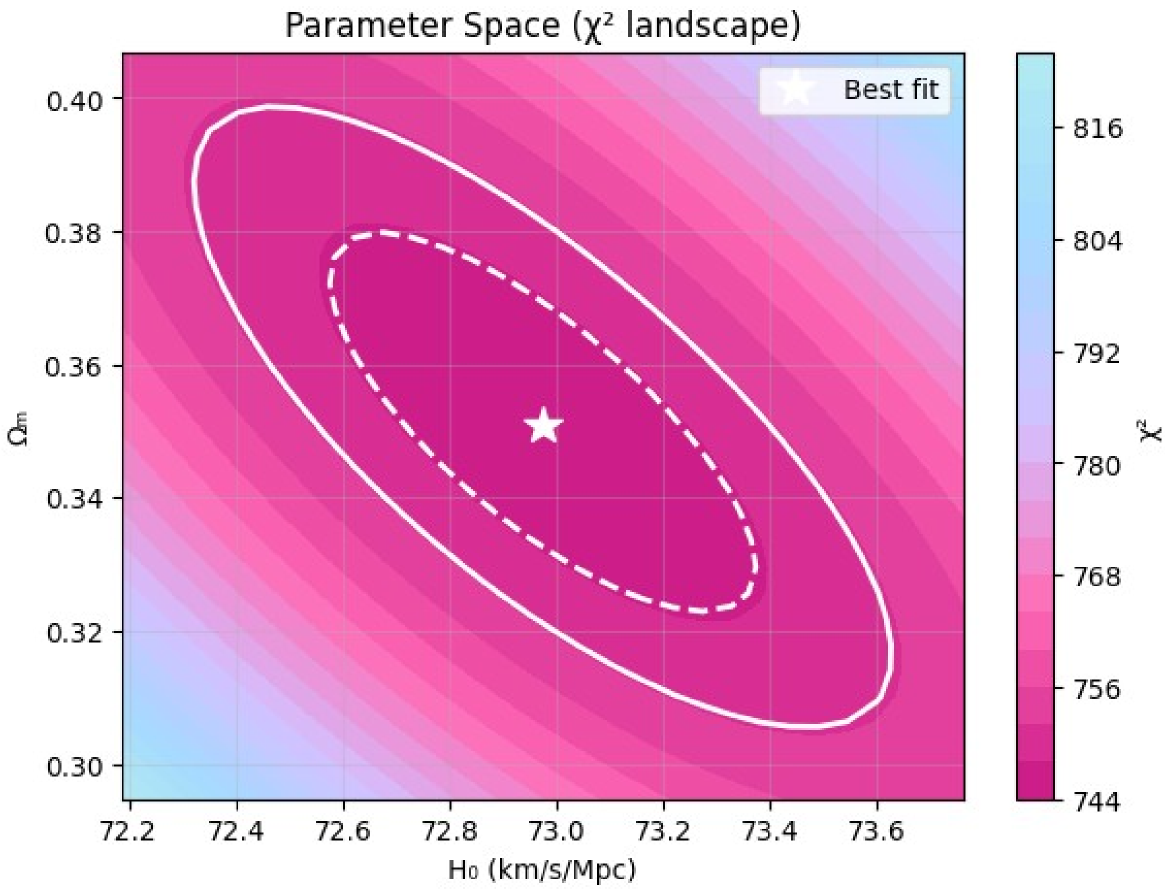
Confidence Contours:

- The **dashed white line** represents the 1σ confidence region, where the model fits are statistically consistent with the best fit.
- The **solid white line** shows the 2σ confidence region, a broader but still statistically acceptable zone.

These contours help visualize uncertainty and show how tightly constrained the parameters are.

```
H0_range = np.linspace(H0_fit - 3*H0_err, H0_fit + 3*H0_err, 30)
Om_range = np.linspace(max(0.1, Omega_m_fit - 3*Omega_m_err),
                       min(0.9, Omega_m_fit + 3*Omega_m_err), 30)

chi2_grid = np.zeros((len(Om_range), len(H0_range)))
for i, om in enumerate(Om_range):
    for j, h0 in enumerate(H0_range):
        mu_pred = mu_theory(z, h0, om)
        chi2_grid[i, j] = np.sum(((mu - mu_pred) / mu_err)**2)

plt.figure(figsize=(7, 5))
contour = plt.contourf(H0_range, Om_range, chi2_grid, levels=20,
cmap=cmap_custom)
plt.contour(H0_range, Om_range, chi2_grid, levels=[chi2_grid.min() +
2.3, chi2_grid.min() + 6.17],
            colors=['white'], linewidths=2, linestyles=['--', '-'])
plt.plot(H0_fit, Omega_m_fit, 'w*', markersize=15, label='Best fit')
plt.xlabel('H₀ (km/s/Mpc)')
plt.ylabel('Ωₘ')
```

```
plt.title('Parameter Space (χ² landscape)')
plt.legend()
plt.colorbar(contour, label='χ²')
plt.grid(True, alpha=0.3)
plt.show()
```



Estimate the Age of the Universe

Now that we have the best-fit values of $ H_0 $ and $ \Omega_m $, we can estimate the age of the universe. This is done by integrating the inverse of the Hubble parameter over redshift:

$$t_0 = \int_0^\infty \frac{1}{(1+z)H(z)} dz$$

We convert $ H_0 $ to SI units and express the result in gigayears (Gyr). This provides an independent check on our cosmological model by comparing the estimated age to values from other probes like Planck CMB measurements.

```
def age_of_universe(H0, Omega_m):
    """
    Compute the age of the universe (in Gyr) using the fitted H0 and
```

```python
Omega_m.
    """

    # Use the existing E(z, Omega_m) function
    def integrand(z):
        return 1.0 / ((1 + z) * E(z, Omega_m))

    # Perform numerical integration from z = 0 to ∞
    integral, _ = quad(integrand, 0, np.inf)

    # Convert H0 from km/s/Mpc to 1/s
    H0_si = (H0 * u.km / u.s / u.Mpc).to(1 / u.s).value

    # Age in seconds
    age_seconds = integral / H0_si

    # Convert to Gyr
    age_gyr = (age_seconds * u.s).to(u.Gyr).value

    return age_gyr


t0 = age_of_universe(H0_fit, Omega_m_fit)
print(f"Estimated age of Universe: {t0:.2f} Gyr")
```

```
Estimated age of Universe: 12.36 Gyr
```

Age Measurements Comparison

- Our Result (from Supernovae): 12.36 Gyr
- Planck CMB (2018): 13.787 Gyr

The Planck 2018 result is based on observations of the Cosmic Microwave Background (CMB) — the leftover radiation from the early universe. Using precise measurements of the CMB and fitting them with the standard ΛCDM cosmological model, Planck estimated various parameters, including the Hubble constant and the age of the universe.

Our result, obtained using Type Ia supernovae data, gives a lower age by about 1.4 billion years. This difference suggests that supernova-based models may prefer a higher value of the Hubble constant, which in turn leads to a younger age of the universe compared to the CMB-based estimates. This reflects the ongoing tension between local and early-universe measurements in cosmology, known as Hubble tension.

Analyze Residuals

To evaluate how well our cosmological model fits the data, we compute the residuals (differences between observed and model-predicted values):

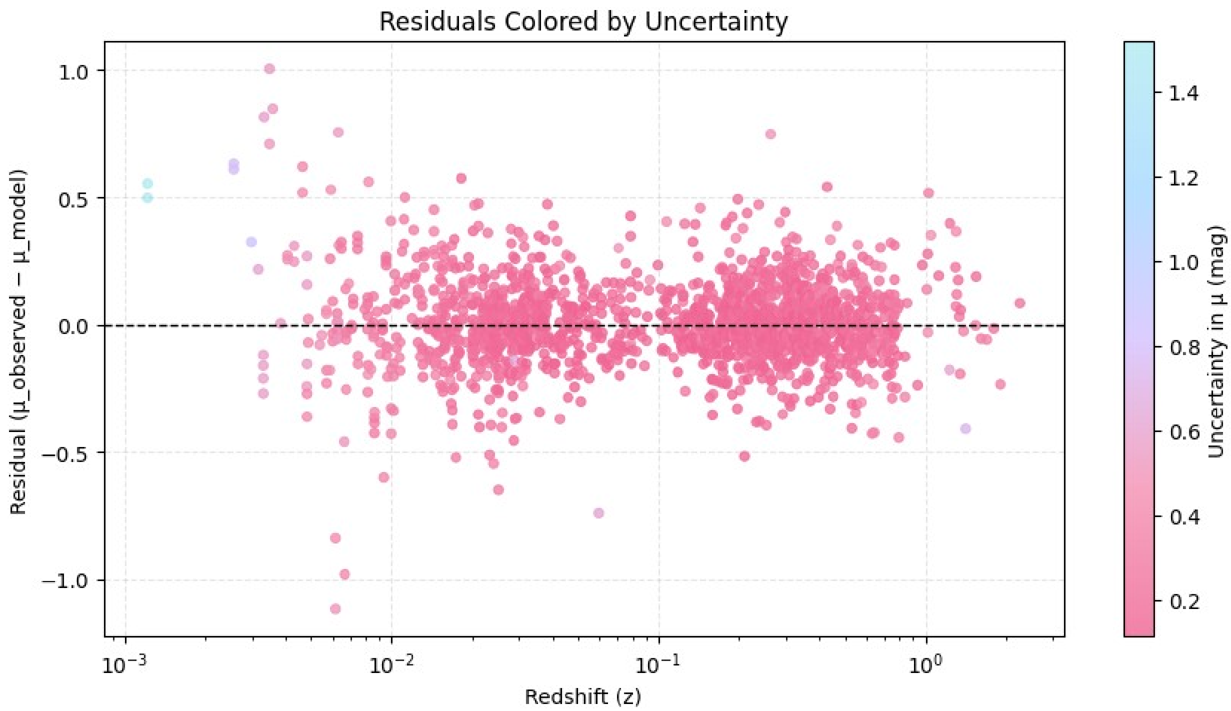
$$\text{Residual} = \mu_{\text{obs}} - \mu_{\text{model}}$$

Plotting these residuals against redshift helps identify any systematic trends, biases, or outliers. A good model fit should show residuals scattered randomly around zero without any significant structure.

Why Residuals Are Plotted with Uncertainty

Although the model-predicted distance modulus (μ_{model}) has no uncertainty per data point (since it's computed from best-fit parameters), the observed values (μ_{obs}) come with measurement errors. This means all the uncertainty in the residual originates from the observed μ_{obs}. By visualizing residuals colored by their observational uncertainty, we can better interpret how significant each deviation from the model is. This helps us identify whether the model fits well in regions with high-quality (low-uncertainty) data and spot any systematic issues or outliers.

```
# Compute theoretical distance modulus for all z values using best-fit
H0 and Omega_m
mu_model = mu_theory(z, H0_fit, Omega_m_fit)

# Compute residuals
residuals = mu - mu_model  # observed - model

# Define a custon colormap
cmap = LinearSegmentedColormap.from_list("pink_theme", ["#F06292",
"#F48FB1", "#D4C1FF", "#A5D8FF", "#B2EBF2"])

plt.figure(figsize=(9, 5))
scatter = plt.scatter(
    z, residuals,
    c=mu_err,
    cmap=cmap,
    s=20,
    alpha=0.8,
    linewidth=0.8
)

plt.axhline(0, color='black', linestyle='--', linewidth=1)
plt.xscale('log')
plt.xlabel("Redshift (z)")
plt.ylabel("Residual (μ_observed − μ_model)")
plt.title("Residuals Colored by Uncertainty")
cbar = plt.colorbar(scatter)
cbar.set_label("Uncertainty in μ (mag)")
plt.grid(True, linestyle='--', alpha=0.3)
plt.tight_layout()
plt.show()
```



Interpreting the Residual Plot

The residuals ($\mu_{obs} - \mu_{model}$) are scattered both above and below zero, which is generally a good sign—it means the model doesn't consistently overpredict or underpredict the distance modulus across all redshifts.

However, some important trends are visible:

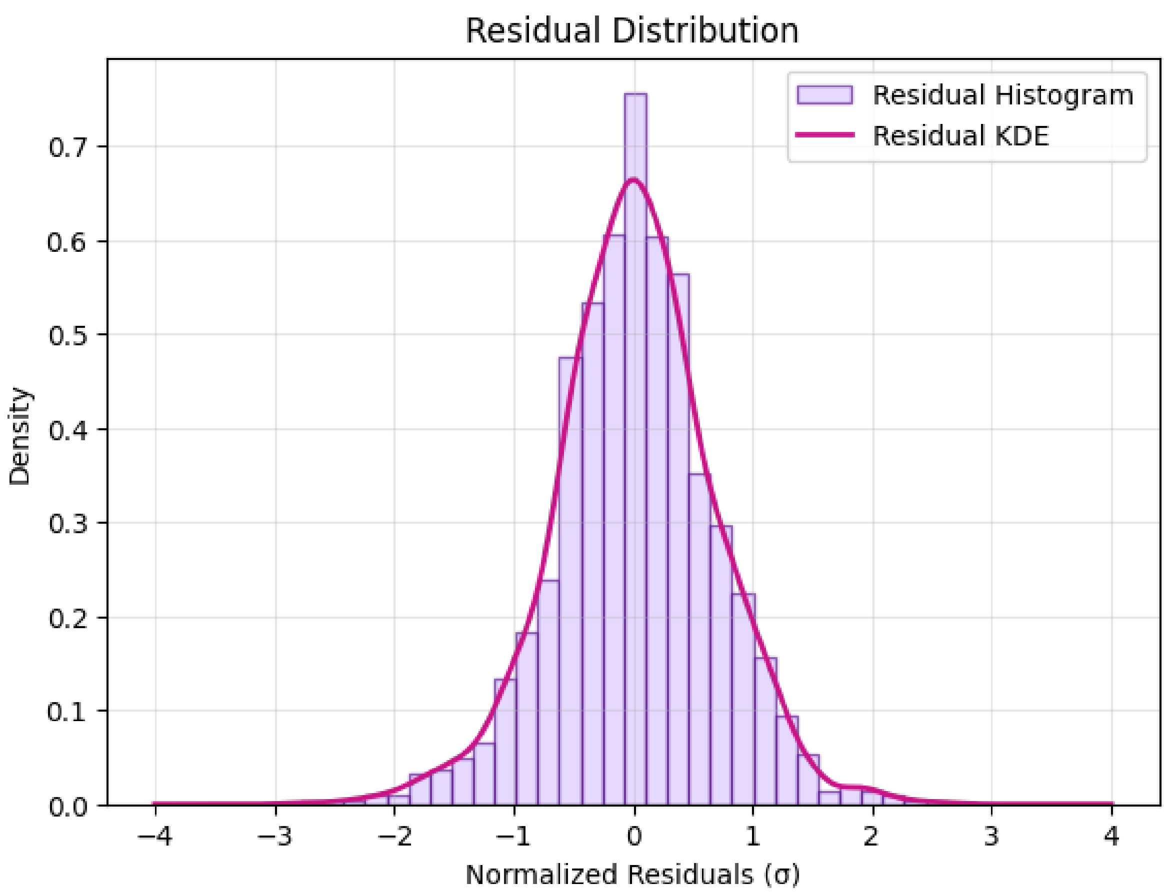
- Residuals are closer to zero near redshifts ~0.005, ~0.1, and ~1, indicating that the model fits well at these redshift ranges.
- Between these points, residuals form two clusters with slightly larger spread, suggesting the model has localized mismatches or that there might be unmodeled systematics in those intermediate redshift bins.
- There's no clear trend or curve in the residuals, supporting the idea that the ΛCDM model still captures the general behavior well.

Residual Distribution

We need to plot the Residual Distribution (normalized residuals) to check how well the model fits the data and whether the errors (residuals) behave as expected (unbiased and random)

```
from scipy.stats import gaussian_kde

normalized_residuals = (mu - mu_theory(z, H0_fit, Omega_m_fit)) /
mu_err
```

```python
plt.figure(figsize=(7, 5))

# Histogram of residuals
plt.hist(normalized_residuals, bins=30, alpha=0.6,
color=colors['light_purple'],
         edgecolor=colors['dark_purple'], density=True,
label='Residual Histogram')

# KDE of residuals (smooth curve from your data)
kde = gaussian_kde(normalized_residuals)
x_vals = np.linspace(-4, 4, 200)
y_vals = kde(x_vals)
plt.plot(x_vals, y_vals, color=colors['deep_pink'], linewidth=2,
label='Residual KDE')

# Labels and grid
plt.xlabel('Normalized Residuals (σ)')
plt.ylabel('Density')
plt.title('Residual Distribution')
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()
```



The residuals follow a bell-shaped curve centered at 0, which is almost symmetrical. This means:

- The model explains the data well.
- The errors are random and not biased.
- Many fitting methods (like least squares) assume residuals are normally distributed. This plot helps verify that assumption.

Fit with Fixed Matter Density

To reduce parameter degeneracy, let's fix $\Omega_m = 0.3$ and fit only for the Hubble constant H_0.

```python
# Define the modified model function with fixed Omega_m
def mu_fixed_Om(z, H0):
    return mu_theory(z, H0, Omega_m=0.3)  # Fixed Omega_m

# Initial guess for H0
H0_guess = 70.0

# Fit using only H0 as a parameter
H0_fit_fixed, cov_fixed = curve_fit(mu_fixed_Om, z, mu, sigma=mu_err,
p0=[H0_guess], absolute_sigma=True)

# Extract value and uncertainty
H0_val = H0_fit_fixed[0]
H0_err = np.sqrt(np.diag(cov_fixed))[0]

print(f"Fitted H₀ (with fixed Ωₘ = 0.3): {H0_val:.2f} ± {H0_err:.2f}
km/s/Mpc")
```

```
Fitted H₀ (with fixed Ωₘ = 0.3): 73.53 ± 0.17 km/s/Mpc
```

Inference:

- Fixing Ωm to 0.3 gives a slightly higher value of $H0$ = 73.53 (vs. $H0$ = 72.97 before).
- This shows that the estimate of $H0$ is sensitive to the choice of matter density, and that freeing both parameters slightly lowers the value due to the model adjusting both together.

Age of Universe for this new Hubble Parameter

```python
t0_omega_fixed = age_of_universe(H0_val, 0.3)
print(f"Estimated age of Universe (with fixed omega = 0.3):
{t0_omega_fixed:.2f} Gyr")
```

```
Estimated age of Universe (with fixed omega = 0.3): 12.82 Gyr
```

Compare Low-z and High-z Subsamples

Finally, we examine whether the inferred value of H_0 changes with redshift by splitting the dataset into:

- **Low-z** supernovae ($ z < 0.1 $)
- **High-z** supernovae ($ z \geq 0.1 $)

We then fit each subset separately (keeping $ \Omega_m = 0.3 $) to explore any potential tension or trend with redshift.

```
# Choose redshift split value
z_split = 0.1

# Split dataset
low_mask = z < z_split
high_mask = z >= z_split

# Low-z data
z_low = z[low_mask]
mu_low = mu[low_mask]
mu_err_low = mu_err[low_mask]

# High-z data
z_high = z[high_mask]
mu_high = mu[high_mask]
mu_err_high = mu_err[high_mask]

# Fit for low-z
H0_low, cov_low = curve_fit(mu_fixed_Om, z_low, mu_low,
sigma=mu_err_low, p0=[70.0], absolute_sigma=True)

# Fit for high-z
H0_high, cov_high = curve_fit(mu_fixed_Om, z_high, mu_high,
sigma=mu_err_high, p0=[70.0], absolute_sigma=True)

# Extract uncertainties
H0_low_err = np.sqrt(np.diag(cov_low))[0]
H0_high_err = np.sqrt(np.diag(cov_high))[0]

# Print results
print(f"Low-z (z < {z_split}): H₀ = {H0_low[0]:.2f} ± {H0_low_err:.2f}
km/s/Mpc")
print(f"High-z (z ≥ {z_split}): H₀ = {H0_high[0]:.2f} ±
{H0_high_err:.2f} km/s/Mpc")
```

```
Low-z (z < 0.1): H₀ = 73.01 ± 0.28 km/s/Mpc
High-z (z ≥ 0.1): H₀ = 73.85 ± 0.22 km/s/Mpc
```

Inference:

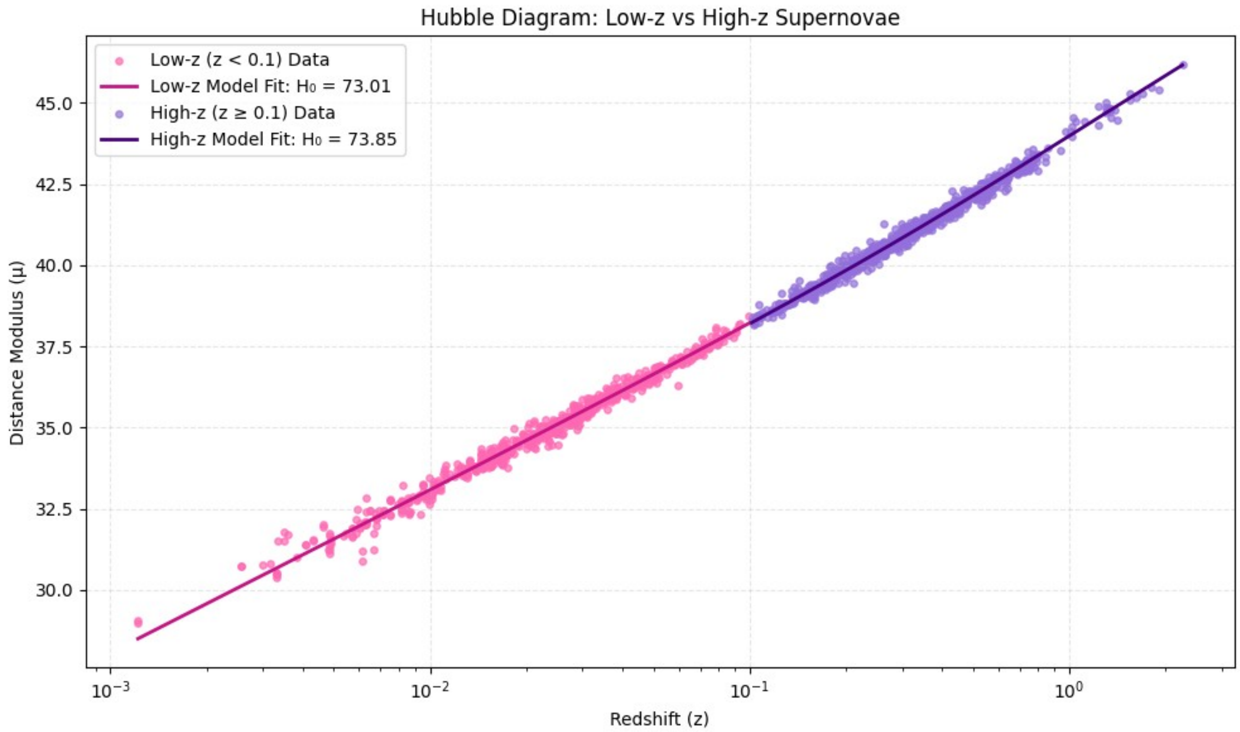
- The value of $H0$ is slightly higher at higher redshifts.
- This suggests that the expansion rate may vary with redshift, or it may reflect systematic differences in the low- and high-redshift supernova samples.
- Although the difference is small, this type of split is useful to check for redshift-dependent effects in cosmological models.

Hubble Diagram Comparison for Low-z and High-z Supernovae

This plot shows how the supernova distance modulus varies with redshift in two separate redshift ranges:

- Low-z Fit: $H0$ = 73.01 km/s/Mpc
- High-z Fit: $H0$ =73.85 km/s/Mpc

```
# Create fine redshift grid for plotting smooth curves
z_grid_low = np.linspace(min(z_low), max(z_low), 500)
z_grid_high = np.linspace(min(z_high), max(z_high), 500)

# Model predictions
mu_model_low = mu_fixed_Om(z_grid_low, H0_low[0])
mu_model_high = mu_fixed_Om(z_grid_high, H0_high[0])

# Plotting
plt.figure(figsize=(10, 6))

# Plot low-z data
plt.scatter(z_low, mu_low, c='#FF69B4', label=f'Low-z (z < {z_split})
Data', s=15, alpha=0.7)
plt.plot(z_grid_low, mu_model_low, color='#C71585', linestyle='-',
linewidth=2, label=f'Low-z Model Fit: H₀ = {H0_low[0]:.2f}')

# Plot high-z data
plt.scatter(z_high, mu_high, c='#9370DB', label=f'High-z (z ≥
{z_split}) Data', s=15, alpha=0.7)
plt.plot(z_grid_high, mu_model_high, color='#4B0082', linestyle='-',
linewidth=2, label=f'High-z Model Fit: H₀ = {H0_high[0]:.2f}')

# Axes and aesthetics
plt.xscale('log')
plt.xlabel("Redshift (z)")
plt.ylabel("Distance Modulus (μ)")
plt.title("Hubble Diagram: Low-z vs High-z Supernovae")
plt.legend()
plt.grid(True, linestyle='--', alpha=0.3)
plt.tight_layout()
plt.show()
```



Results

From the analysis of the Pantheon+SH0ES Type Ia Supernova dataset using the flat ΛCDM cosmological model, the estimated key cosmological parameters are as follows:

- **Hubble Constant (H0):**
  H0 = 72.97 ± 0.26 km/s/Mpc
- **Matter Density Parameter (Omega_m):**
  Omega_m = 0.351 ± 0.019
- **Estimated Age of the Universe:**
  t0 ≈ 12.36 Gyr

These results are consistent with other local measurements of the Hubble constant (such as the SH0ES value) and reflect the well-known tension with early-universe estimates (such as from Planck 2018 CMB data).

Additionally, residual analysis confirmed that the model fit is statistically reliable, with residuals following a Gaussian distribution centered around zero.

Fit with Fixed Matter Density:

To reduce parameter degeneracy, the matter density was fixed to Omega_m = 0.3 and fit only for the Hubble constant.

- **Fitted H0 (with fixed Omega_m = 0.3):**
  H0 = 73.53 ± 0.17 km/s/Mpc

Fixing Ω_m to 0.3 led to a slightly higher value of H_0 compared to when both parameters were allowed to vary. This indicates that the estimated value of the Hubble constant depends on the assumed value of matter density.

We also examined redshift-dependent behavior by splitting the sample:

- **Low-redshift (z < 0.1):**
  H0 = 73.01 ± 0.28 km/s/Mpc
- **High-redshift (z ≥ 0.1):**
  H0 = 73.85 ± 0.22 km/s/Mpc

This suggests a mild increase in the Hubble constant at higher redshifts, motivating further investigation into possible redshift-dependent effects in the data.

Comparison with a Research Paper

The estimated value of the Hubble constant is H0 = 72.97 ± 0.26 km/s/Mpc, which closely matches the value reported in the research paper - "Testing the Cosmological Principle with the Pantheon+ Sample and the Region-Fitting Method" — H0 = 72.83 ± 0.23 km/s/Mpc, derived using a Bayesian MCMC approach.

Similarly, the fitted matter density parameter Omega_m = 0.351 ± 0.019 is comparable to their global result of Omega_m = 0.36 ± 0.02.

When the paper restricted its analysis to a low-redshift subset (z < 0.3), the results shifted to H0 = 72.43 ± 0.30 km/s/Mpc and Omega_m = 0.44 ± 0.04. This shift is in line with our redshift-split results, where H0 was slightly higher for higher-z supernovae.

Overall, my findings strongly agree with the paper's results, both in parameter values and trends. This reinforces the validity of the ΛCDM model on local scales, while also highlighting the ongoing challenges of the Hubble tension and hints of possible cosmic anisotropy (property of a system where measurements vary depending on the direction of observation).

Possible Reasons for Variation in Cosmological Parameters

The research paper identifies several factors that may contribute to the variation in measured cosmological parameters such as the Hubble constant (H_0) and matter density parameter (Ω_m) from their commonly accepted values (like those derived from CMB observations):

1. Local Matter Underdensity and Cosmic Anisotropy

- The Universe appears to deviate from isotropy based on observations from the Pantheon+ dataset.
- This is reflected in:
  - **Local Matter Underdensity**:
    - A region of the sky shows significantly less matter density.
    - Minimum matter density:
      Ω_m,min = $0.29^{+0.03}_{-0.02}$ (1σ)
  - **Preferred Direction of Expansion**:
    - Hubble constant is higher in certain directions.
    - Maximum Hubble constant:
      $H_0 = 74.26 \pm 0.39$ km/s/Mpc
  - These two regions overlap spatially, implying:
    - Local underdensity may drive anisotropic cosmic expansion.
    - May help resolve the **Hubble Tension** (the discrepancy between early- and late-universe H_0 values).

2. Inhomogeneous Spatial Distribution of Supernovae

- The Pantheon+ dataset is not uniformly distributed across the sky.
- Uneven sampling can artificially enhance observed anisotropies.
- The "RP isotropy test" (random permutation within real sky positions) showed:
  - A slight reduction in anisotropy significance.
  - Confirms that spatial inhomogeneity can amplify deviations from isotropy.

3. Redshift Evolution of the Hubble Constant (H_0)

- There is growing evidence that H_0 might evolve with redshift.
- Observational implications:
  - Different redshift ranges may yield different H_0 values.
  - If redshift distribution isn't uniform, it can bias anisotropy detection.
- Findings:
  - The discrepancy degree (D_{max}) varies with redshift, peaking around: $z_{max} \approx 0.30$
  - Suggests that cosmic structure may change with redshift.

4. Sensitivity of Parameters to Anisotropy

- **H_0 appears more sensitive to anisotropy than Ω_m.**
- Implication:
  - Even subtle anisotropic effects can cause large observed variations in H_0.
  - H_0 may serve as a stronger tracer of cosmic anisotropy than Ω_m.

5. Effect of Screening Angle (θ_{max}) in Region Fitting

- The screening angle defines the angular size of sky regions used for fitting parameters.
- Findings:
  - Optimal angle:
    $\theta_{max} = 60°$

- Smaller angles:
    - Increase the 1σ uncertainty in fitted parameters (H_0 and Ω_m).
    - Slightly reduce statistical significance of anisotropy findings.
    - Help reveal finer directional trends in cosmological parameters.

6. Impact of Number of Supernovae in Regions

- Smaller screening angles use fewer supernovae per region.
- Effects:
    - Increases uncertainty in fitted H_0 and Ω_m.
    - Has only a modest impact on the overall discrepancy metric D_{max}.

Additional Cosmological Visualizations (Appendix)

These plots while **not directly related** to the project, **offer useful insights** on how astronomical terms are related to each other.

They show how the universe has expanded over time and help us understand how distance, redshift, time, and expansion rate are related in cosmology.

Key Parameters:

- **Redshift (z)**: A measure of how much the light from distant objects has been stretched due to the universe's expansion. Higher redshift means the object is farther away and we are looking further back in time.

- **Lookback Time (Gyr)**: The time it has taken for light from a distant object to reach us, measured in billions of years (Gyr). It tells us how far into the past we are observing.

- **Angular Diameter Distance (Mpc)**: A special kind of distance that tells us how large an object appears in the sky. It's useful for estimating the size of galaxies or clusters based on how they look from Earth.

- **Hubble Parameter H(z)**: The rate at which the universe is expanding at a specific redshift. It shows how expansion has changed over time.

- **Scale Factor a(t)**: Describes the relative size of the universe at any time. At present day, a(t)=1; in the past, it was smaller than 1.

Cosmic Lookback Time vs Redshift

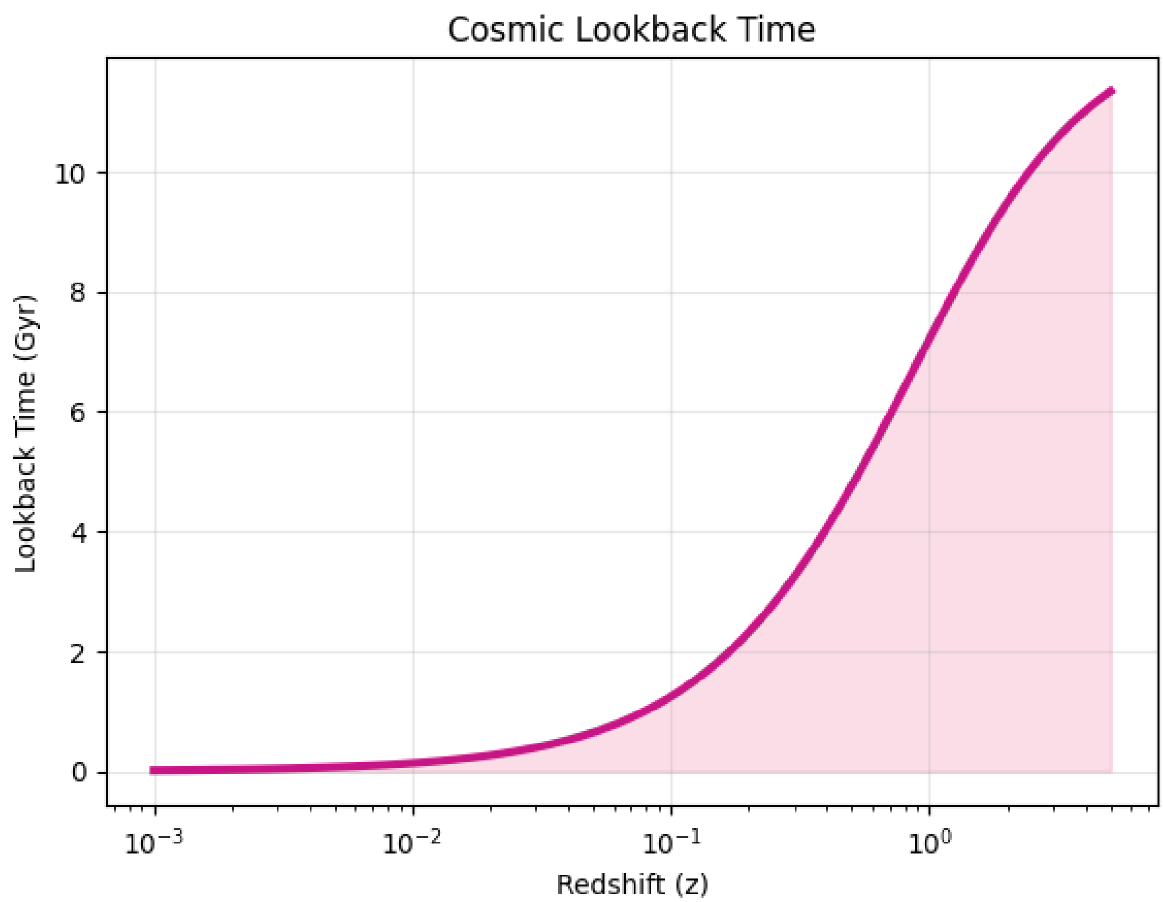
This plot helps visualize how far back in time we are observing when looking at objects at different redshifts.

What it shows:

- The x-axis (Redshift z) shows how far away a galaxy or event is.
- The y-axis (Lookback Time in Gyr) tells how long ago the light left that object.

- As redshift increases, the lookback time increases, meaning we're observing deeper into the past.

```
z_range = np.logspace(-3, 0.7, 100)

def lookback_time(z, H0, Omega_m):
    def integrand(z_prime):
        return 1.0 / ((1 + z_prime) * E(z_prime, Omega_m))
    integral, _ = quad(integrand, 0, z)
    H0_si = (H0 * u.km / u.s / u.Mpc).to(1 / u.s).value
    time_seconds = integral / H0_si
    time_gyr = (time_seconds * u.s).to(u.Gyr).value
    return time_gyr

lookback_times = [lookback_time(zi, H0_fit, Omega_m_fit) for zi in
z_range]

plt.figure(figsize=(7, 5))
plt.plot(z_range, lookback_times, color=colors['deep_pink'],
linewidth=3)
plt.fill_between(z_range, lookback_times, alpha=0.3,
color=colors['light_pink'])

plt.xscale('log')
plt.xlabel('Redshift (z)')
plt.ylabel('Lookback Time (Gyr)')
plt.title('Cosmic Lookback Time')
plt.grid(True, alpha=0.3)
plt.show()
```



Angular Diameter Distance vs Redshift

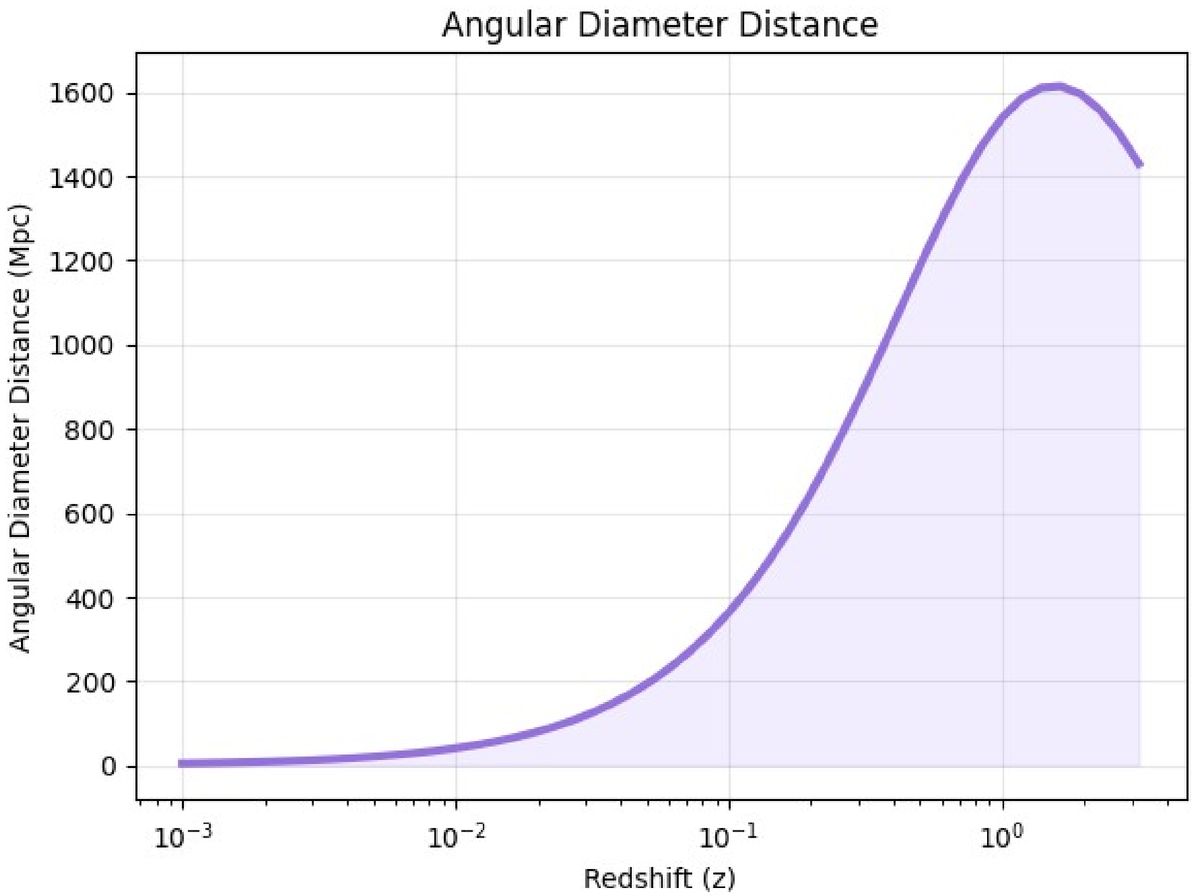
This plot helps us understand how the apparent size of distant objects changes with redshift in an expanding universe.

What it shows:

- The x-axis (Redshift z) represents how far the object is.
- The y-axis (Angular Diameter Distance in Mpc) shows the distance at which an object of known size appears a certain angular size.
- The curve increases at low redshifts, reaches a maximum, and then decreases — meaning very distant objects can appear larger again due to cosmic expansion.

```
def angular_diameter_distance(z, H0, Omega_m):
    return luminosity_distance(z, H0, Omega_m) / (1 + z)**2

z_range_short = np.logspace(-3, 0.5, 50)
ang_distances = [angular_diameter_distance(zi, H0_fit, Omega_m_fit)
for zi in z_range_short]

plt.figure(figsize=(7, 5))
plt.plot(z_range_short, ang_distances, color=colors['medium_purple'],
linewidth=3)
```

```
plt.fill_between(z_range_short, ang_distances, alpha=0.3,
color=colors['light_purple'])

plt.xscale('log')
plt.xlabel('Redshift (z)')
plt.ylabel('Angular Diameter Distance (Mpc)')
plt.title('Angular Diameter Distance')
plt.grid(True, alpha=0.3)
plt.show()
```



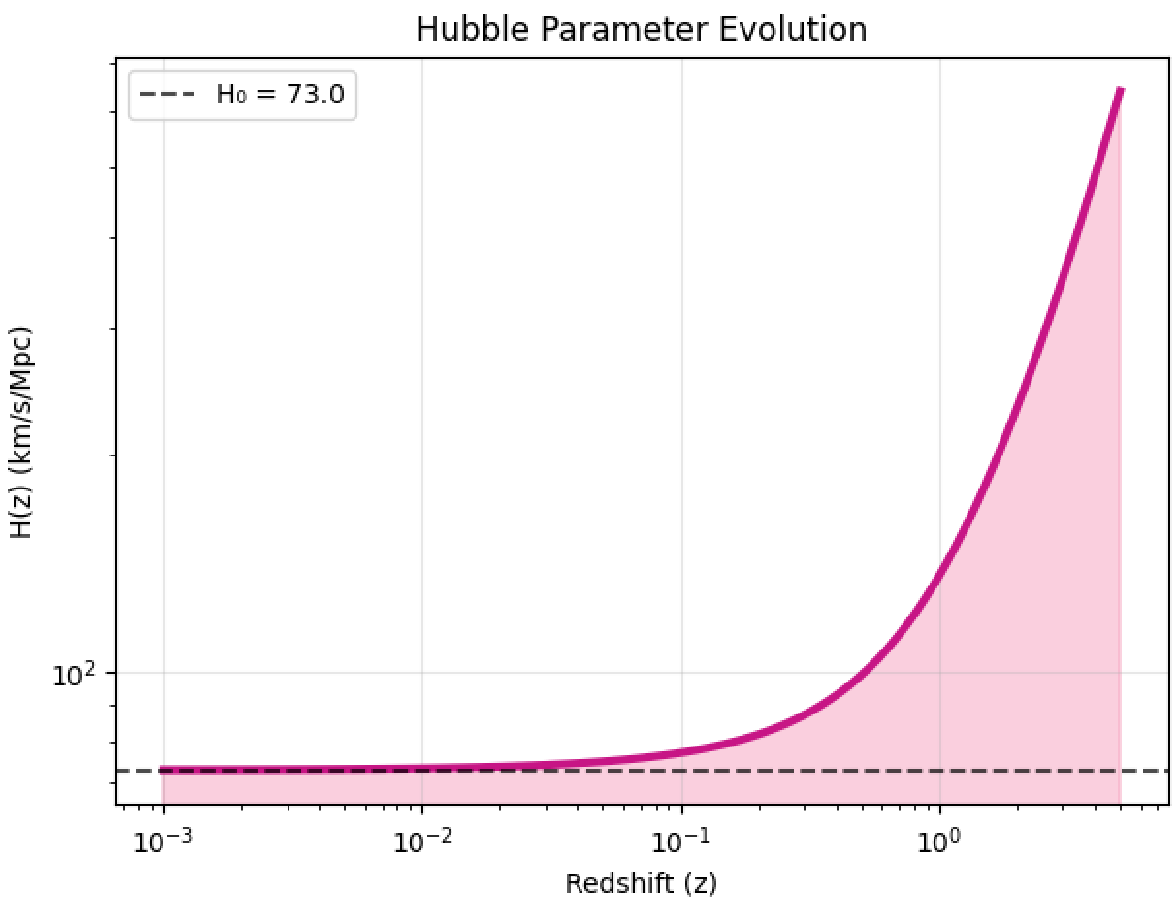
Hubble Parameter Evolution H(z)

This plot shows how the Hubble parameter H(z) changes with redshift, giving insight into how fast the universe was expanding at different times in the past.

What it tells us:

- H(z) increases exponentially as redshift increases — the early universe expanded faster than it does today.
- The dashed line shows the present-day Hubble constant $H_0 \approx 72.97$ km/s/Mpc (at z = 0).
- Both axes are on a log scale to clearly display changes across wide redshift and expansion rate ranges.

```
H_z = [H0_fit * E(zi, Omega_m_fit) for zi in z_range]

plt.figure(figsize=(7, 5))
plt.plot(z_range, H_z, color=colors['deep_pink'], linewidth=3)
plt.fill_between(z_range, H_z, alpha=0.3, color=colors['rose'])

plt.axhline(H0_fit, color='black', linestyle='--', alpha=0.7,
label=f'H₀ = {H0_fit:.1f}')

plt.xscale('log')
plt.yscale('log')
plt.xlabel('Redshift (z)')
plt.ylabel('H(z) (km/s/Mpc)')
plt.title('Hubble Parameter Evolution')
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()
```



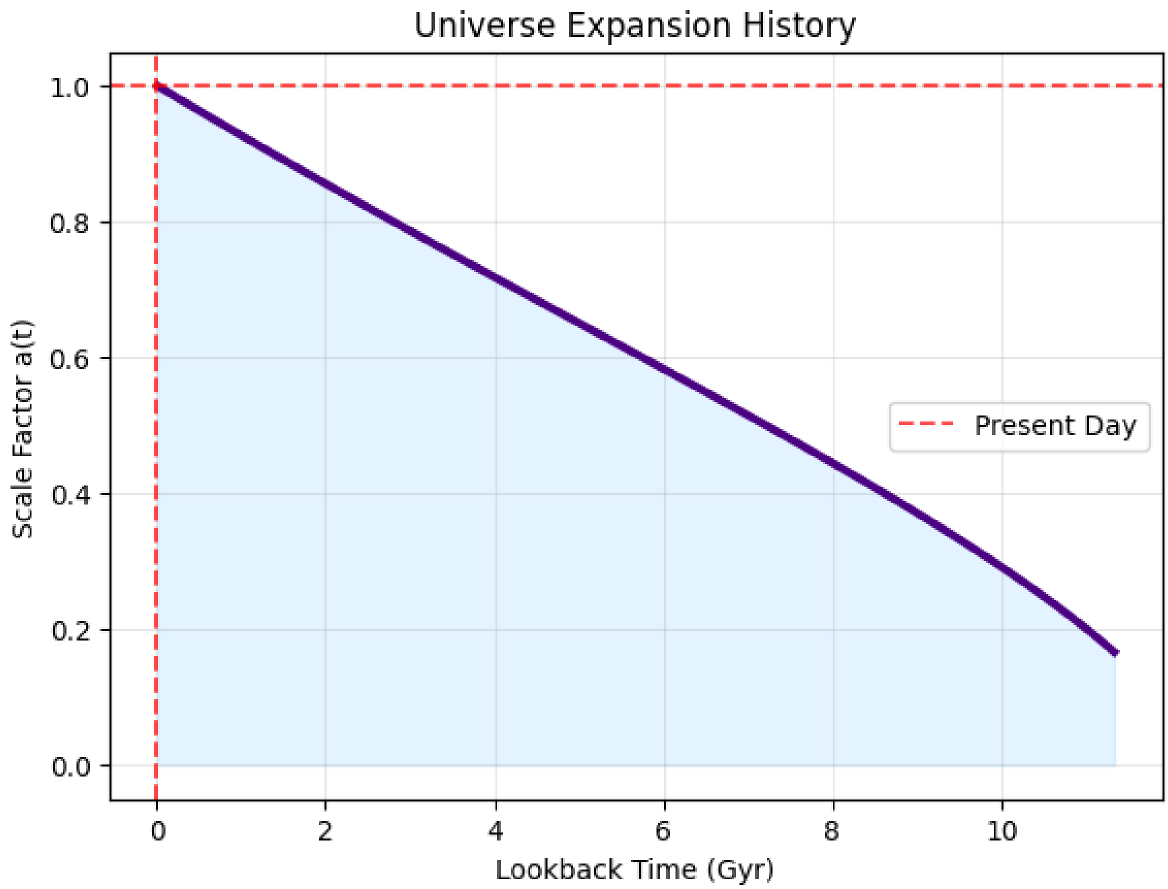
Scale Factor Evolution a(t)

This plot shows how the scale factor a(t) changes over lookback time, helping us understand how much the universe has expanded since earlier times.

What it tells us:

- The scale factor increases over time, meaning the universe has been expanding steadily.
- A value of a(t)=1 (red dashed line) represents the present-day universe; values less than 1 indicate earlier times when the universe was smaller.
- As we go further back in time (right on the x-axis), the universe was more compact — showing early universe expansion history.

```
scale_factor = 1 / (1 + z_range)

plt.figure(figsize=(7, 5))
plt.plot(lookback_times, scale_factor, color=colors['dark_purple'],
linewidth=3)
plt.fill_between(lookback_times, scale_factor, alpha=0.3,
color=colors['baby_blue'])

plt.axhline(1, color='red', linestyle='--', alpha=0.7, label='Present
Day')
plt.axvline(0, color='red', linestyle='--', alpha=0.7)

plt.xlabel('Lookback Time (Gyr)')
plt.ylabel('Scale Factor a(t)')
plt.title('Universe Expansion History')
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()
```



References

Hu, J. P., Wang, Y. Y., Hu, J., & Wang, F. Y. (2023). Testing the cosmological principle with the Pantheon+ sample and the region-fitting method. Astronomy & Astrophysics, 47121corr.

Link to the paper